A Strong CP Primer

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ABSTRACT: The aim of this lecture is to understand the strong CP problem [1–4] and its possible solutions. Here I assume the students have basic knowledge of quantum field theory such as

- Symmetry and Noether current
- Spontaneous symmetry breaking and Nambu–Goldstone boson
- Chiral anomaly and Fujikawa method
- Lagrangian and matter content of the Standard Model

Contents

0 Notation

Notation in this lecture note is the same as Peskin and Schroeder's textbook. The metric is mostly minus, $g_{\mu\nu} = \text{diag}(1,-1,-1,-1)$. The Levi-Civita tensor is defined as $\epsilon^{0123} = -\epsilon_{0123} = 1$. We take the following representation of gamma matrices:

$$\gamma^0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad \gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -I_2 & 0 \\ 0 & I_2 \end{pmatrix}. \tag{0.1}$$

The projection operators are defined as

$$P_L = \frac{1}{2}(1-\gamma^5), \qquad P_R = \frac{1}{2}(1+\gamma^5). \tag{0.2}$$

1 Why is neutron electric dipole moment so small?

Interaction between a particle with nonzero spin and static electromagnetic field in the vacuum is described by the following effective Hamiltonian:

$$H = -\mu\frac{\vec{s}\cdot\vec{B}}{|\vec{s}|} - d\frac{\vec{s}\cdot\vec{E}}{|\vec{s}|}, \tag{1.1}$$

where $\vec{s}$ is the spin operator of the particle. The coefficient μ and d are called magnetic dipole moment and electric dipole moment. Both of them have the same mass dimension, (charge) $\times$ (length).

For the neutron, the observed value of μ_n is [5]

$$\mu_n = -1.9 \times \frac{e\hbar}{2m_p} \simeq -2.0\times 10^{-14}\ e\,\text{cm}. \tag{1.2}$$

On the other hand, the most precise measurement on d_n is given by the PSI experiment as [6]

$$d_n = (0.0 \pm 1.1_{\text{stat}} \pm 0.2_{\text{sys}}) \times 10^{-26}\ e\,\text{cm}. \tag{1.3}$$

This result can be interpreted as an upper bound on d_n on 90 % C.L.:

$$|d_n| < 1.8\times 10^{-26}\ e\,\text{cm}. \tag{1.4}$$

We can see that the ratio of two dipole moments μ_n and d_n is bounded as

$$\left|\frac{d_n}{\mu_n}\right| \lesssim 10^{-12}. \tag{1.5}$$

We can see that d_n is quite suppressed compared to μ_n. Why so small!?

Smallness of d_n could be explained by the existence of symmetry to forbid d_n. There are two natural candidates of such a symmetry:

	$\vec{s}$	$\vec{E}$	$\vec{B}$	μ	d
P	$+$	$-$	$+$	$+$	$-$
T	$-$	$+$	$-$	$+$	$-$

Table 1. Symmetry and dipole moments.

- Parity symmetry (P) : $(t, \vec{x}) \to (t, -\vec{x})$
- Time reversal symmetry (T) : $(t, \vec{x}) \to (-t, \vec{x})$

Let us see how $\vec{E}$ and $\vec{B}$ should behave under P and T transformation from Maxwell equations:

$$\vec{\nabla} \cdot \vec{B} = 0, \qquad \vec{\nabla} \times \vec{E} + \frac{\partial}{\partial t}\vec{B} = \vec{0}, \tag{1.6}$$

$$\vec{\nabla} \cdot \vec{E} = \rho, \qquad \vec{\nabla} \times \vec{B} - \frac{\partial}{\partial t}\vec{E} = \vec{j}. \tag{1.7}$$

P and T transformation for ρ and $\vec{j}$ is

$$P: \qquad \rho(t,\vec{x}) \to \rho(t,-\vec{x}), \qquad \vec{j}(t,\vec{x}) \to -\vec{j}(t,-\vec{x}), \tag{1.8}$$

$$T: \qquad \rho(t,\vec{x}) \to \rho(-t,\vec{x}), \qquad \vec{j}(t,\vec{x}) \to -\vec{j}(-t,\vec{x}). \tag{1.9}$$

Thus, $\vec{E}$ and $\vec{B}$ behaves as

$$P: \qquad \vec{E}(t,\vec{x}) \to -\vec{E}(t,-\vec{x}), \qquad \vec{B}(t,\vec{x}) \to \vec{B}(t,-\vec{x}), \tag{1.10}$$

$$T: \qquad \vec{E}(t,\vec{x}) \to \vec{E}(-t,\vec{x}), \qquad \vec{B}(t,\vec{x}) \to -\vec{B}(-t,\vec{x}). \tag{1.11}$$

Thus, under P, $\vec{E}$ is odd and $\vec{B}$ is even. Under T, $\vec{E}$ is even and $\vec{B}$ is odd.[1] The behavior of $\vec{s}$ is same as the angular momentum, so it is even for P and odd for T. See table 1.

Thus, we can immediately see that, if we impose either P or T symmetry on the Hamiltonian, d_n should be zero but nonzero μ_n is allowed. So do we have either P or T symmetry in nature? Actually, the situation is more complicated as we will see in this lecture.

In section 2, we briefly review the most generic Lagrangian of QED and QCD, and also their symmetries. In section 3, we review the BPST instanton and see that θ-term in non-abelian gauge theory cannot be neglected in general. In section 4, we introduce effective field theory for mesons to describe the physical consequences of θ-term, and we will see the connection between $U(1)_A$ problem and the strong CP problem. In section

[1]Note that the scalar potential ϕ and vector potential $\vec{A}$ forms a four-vector $A^\mu = (\phi, \vec{A})$. $\vec{E}$ and $\vec{B}$ are written in terms of A^μ as

$$\vec{E} = -\vec{\nabla}\phi - \partial_t \vec{A} = (-F^{01}, -F^{02}, -F^{03}), \tag{1.12}$$

$$\vec{B} = \vec{\nabla} \times \vec{A} = (-F^{23}, -F^{31}, -F^{12}), \tag{1.13}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. From Maxwell equation $\partial_\mu F^{\mu\nu} = j^\nu$, we can observe that A_μ and j_μ should behave in the same way under P and T transformation, i.e., $A_0 \to A_0$ and $A_i \to -A_i$.

5, we will discuss CP violation in the Standard Model. In section 6, we will reformulate the strong CP problem by using topological susceptibility. In section 7, we describe the QCD axion. In section 8, we describe the Nelson-Barr mechanism which is based on the spontaneous symmetry breaking of CP symmetry.

2 C, P, and CP symmetry

In this section, we briefly review C, P, and CP symmetry in QED and QCD. The following textbook is useful to understand this section.

- Peskin-Schroeder [7] §3, §15, §19

2.1 QED

Let us start our discussion from QED. The Lagrangian of QED is given by

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi, \tag{2.1}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and $D_\mu = \partial_\mu + ieA_\mu$.

A_μ is the photon field and ψ is the electron field. This Lagrangian is invariant under the following gauge transformation:

$$\psi \to \exp\left(-i\alpha(x)\right)\psi, \qquad A_\mu \to A_\mu + \frac{1}{e}\partial_\mu\alpha(x). \tag{2.2}$$

$F_{\mu\nu}$ is invariant under this gauge transformation. Since

$$\partial_\mu\psi \to \exp\left(-i\alpha(x)\right)\left(\partial_\mu\psi - i(\partial_\mu\alpha)\psi\right), \tag{2.3}$$

we can find

$$D_\mu\psi \to \exp\left(-i\alpha(x)\right)D_\mu\psi. \tag{2.4}$$

2.1.1 Global symmetries in QED

Let us investigate global symmetries in QED. The Lagrangian of QED (2.1) is invariant under the following global phase rotation:

$$\psi \to e^{i\alpha}\psi. \tag{2.5}$$

This global $U(1)$ symmetry is associated with the conservation of electron number. Noether current associated with this global $U(1)$ symmetry is given by

$$J^\mu = \bar{\psi}\gamma^\mu\psi. \tag{2.6}$$

This current is conserved as

$$\partial_\mu J^\mu = 0. \tag{2.7}$$

In addition to the global $U(1)$ symmetry, the Lagrangian of QED (2.1) is invariant under C, P, and T transformation.

The P transformation is defined as

$$\psi(t,\vec{x}) \to \gamma^0\psi(t,-\vec{x}), \tag{2.8}$$

$$A_\mu(t,\vec{x}) \to P_\mu^{\ \nu} A_\nu(t,-\vec{x}), \tag{2.9}$$

where $P_\mu^{\ \nu} = \mathrm{diag}(1,-1,-1,-1)$. Under this P transformation, the electron current operator behaves as

$$\bar{\psi}\gamma^0\psi \to (\psi^\dagger\gamma^0\gamma^0)\gamma^0(\gamma^0\psi) = \bar{\psi}\gamma^0\psi, \tag{2.10}$$

$$\bar{\psi}\gamma^i\psi \to (\psi^\dagger\gamma^0\gamma^0)\gamma^i(\gamma^0\psi) = -\bar{\psi}\gamma^i\psi, \tag{2.11}$$

$$\bar{\psi}\psi \to (\psi^\dagger\gamma^0\gamma^0)(\gamma^0\psi) = \bar{\psi}\psi. \tag{2.12}$$

The C transformation is defined as

$$\psi(t,\vec{x}) \to i\gamma^2\psi^*(t,\vec{x}), \tag{2.13}$$

$$A_\mu(t,\vec{x}) \to -A_\mu(t,\vec{x}). \tag{2.14}$$

The electron current operator behaves as

$$\begin{aligned}\bar{\psi}\gamma^\mu\psi &\to \psi^T(i\gamma^2)^\dagger\gamma^0\gamma^\mu(i\gamma^2)\psi^* \\ &= -\psi^\dagger(i\gamma^2)^T(\gamma^\mu)^T(\gamma^0)^T(i\gamma^2)^*\psi \\ &= \psi^\dagger\gamma^2(\gamma^\mu)^T\gamma^0\gamma^2\psi \\ &= -\bar{\psi}\gamma^\mu\psi, \end{aligned}\tag{2.15}$$

$$\begin{aligned}\bar{\psi}\psi &\to \psi^T(i\gamma^2)^\dagger\gamma^0(i\gamma^2)\psi^* \\ &= -\psi^\dagger(i\gamma^2)^T(\gamma^0)^T(i\gamma^2)^*\psi \\ &= \psi^\dagger\gamma^2\gamma^0\gamma^2\psi \\ &= \bar{\psi}\psi. \end{aligned}\tag{2.16}$$

Here we have used $(\gamma^\mu)^T = \gamma^2\gamma^0\gamma^\mu\gamma^0\gamma^2$.

The T transformation is defined as

$$\psi(t,\vec{x}) \to -\gamma^1\gamma^3\psi(-t,\vec{x}), \tag{2.17}$$

$$A_\mu(t,\vec{x}) \to T_\mu^{\ \nu} A_\nu(-t,\vec{x}), \tag{2.18}$$

where $T_\mu^{\ \nu} = \mathrm{diag}(-1,1,1,1)$. Then, ψ is transformed as

$$\bar{\psi}\gamma^\mu\psi \to \bar{\psi}\gamma^3\gamma^1(\gamma^\mu)^*\gamma^1\gamma^3\psi = \begin{cases}\bar{\psi}\gamma^0\psi & (\mu=0) \\ -\bar{\psi}\gamma^i\psi & (\mu=i)\end{cases}, \tag{2.19}$$

$$\bar{\psi}\psi \to \bar{\psi}\gamma^3\gamma^1\gamma^1\gamma^3\psi = \bar{\psi}\psi. \tag{2.20}$$

Here we have used that T transformation is anti-unitary, so $(\gamma^\mu)^*$ appears in the transformation of $\bar{\psi}\gamma^\mu\psi$.

Thus, the Lagrangian given in eq. (2.1) is invariant under C, P, and T transformation. Also the Lagrangian is invariant under CP transformation.

2.1.2 Axial symmetry, chiral anomaly, and Fujikawa method

Let us consider the limit of $m \to 0$ in QED. In this limit, the Lagrangian looks invariant under the following axial phase rotation:

$$\psi \to \psi' = e^{i\alpha\gamma_5}\psi. \tag{2.21}$$

Let us call this symmetry $U(1)_A$ symmetry. Noether current associated with this $U(1)_A$ symmetry is given by

$$J_5^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi. \tag{2.22}$$

Then, we can naively expect that this current is conserved as

$$\partial_\mu J_5^\mu = 0? \tag{2.23}$$

However, this $U(1)_A$ symmetry is broken by quantum effect, chiral anomaly! We can see the chiral anomaly by using Fujikawa method.

The partition function of QED is given by

$$Z = \int \mathcal{D}A \int \mathcal{D}\psi\mathcal{D}\bar{\psi} \exp\left(i\int d^4x\mathcal{L}[\psi, A]\right) \tag{2.24}$$

Let us replace ψ to ψ' as

$$Z = \int \mathcal{D}A \int \mathcal{D}\psi'\mathcal{D}\bar{\psi}' \exp\left(i\int d^4x\mathcal{L}[\psi', A]\right), \tag{2.25}$$

and write ψ' by the following infinitesimal chiral field redefinition:

$$\psi(x) \to \psi'(x) = (1 + i\alpha(x)\gamma_5)\psi(x), \tag{2.26}$$
$$\bar{\psi}(x) \to \bar{\psi}'(x) = \bar{\psi}(x)(1 + i\alpha(x)\gamma_5). \tag{2.27}$$

Then, the Lagrangian $\mathcal{L}[\psi', A]$ can be written in term of ψ as

$$\begin{aligned}\int d^4x\bar{\psi}'(x)(i\gamma^\mu D_\mu)\psi'(x) &= \int d^4x \left[\bar{\psi}(x)(i\gamma^\mu D_\mu)\psi(x) - (\partial_\mu\alpha(x))\bar{\psi}\gamma^\mu\gamma^5\psi\right] \\ &= \int d^4x \left[\bar{\psi}(x)(i\gamma^\mu D_\mu)\psi(x) + \alpha(x)\partial_\mu J_5^\mu\right], \end{aligned} \tag{2.28}$$

where $J_5^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi$ is the axial current The integral measure $\mathcal{D}\psi'\mathcal{D}\bar{\psi}'$ is not same as $\mathcal{D}\psi\mathcal{D}\bar{\psi}$!

$$\mathcal{D}\psi'\mathcal{D}\bar{\psi}' \simeq \mathcal{D}\psi\mathcal{D}\bar{\psi} \exp\left(i\int d^4x\alpha(x)\frac{e^2}{8\pi^2}F_{\mu\nu}\tilde{F}^{\mu\nu}\right), \tag{2.29}$$

$\tilde{F}_{\mu\nu} = (1/2)\epsilon_{\mu\nu\rho\sigma}F^{\rho\sigma}$ is the dual field strength.

Thus, the partition function Z can be written as

$$Z = \int \mathcal{D}A \int \mathcal{D}\psi'\mathcal{D}\bar{\psi}' \exp\left(i\int d^4x\mathcal{L}[\psi', A]\right) = \int \mathcal{D}A \int \mathcal{D}\psi\mathcal{D}\bar{\psi} \exp\left(i\int d^4x(\mathcal{L}[\psi, A] + \delta\mathcal{L})\right), \tag{2.30}$$

where the correction to the Lagrangian is given by

$$\delta\mathcal{L} = \alpha(x)\left[(\partial_\mu J_5^\mu) + \frac{e^2}{8\pi^2}F_{\mu\nu}\tilde{F}^{\mu\nu}\right]. \tag{2.31}$$

As a result, axial current is not conserved in the background of electromagnetic field as

$$\partial_\mu J_5^\mu = -\frac{e^2}{8\pi^2}F_{\mu\nu}\tilde{F}^{\mu\nu}. \tag{2.32}$$

Thus, the $U(1)_A$ symmetry is broken in the electromagnetic background with $F_{\mu\nu}\tilde{F}^{\mu\nu} \neq 0$.

2.1.3 The most generic Lagrangian of QED

We gave the QED Lagrangian in eq. (2.1). However, is this the most generic renormalizable Lagrangian under the gauge symmetry? Let us construct the most generic renormalizable Lagrangian of A_μ and ψ which is invariant under the gauge transformation given in eq. (2.2).

Lagrangian is made of gauge-invariant Lorentz-scalar terms. There is no such term up to dimension 2. Possible dimension 3 operators are

$$\bar{\psi}\psi, \qquad \bar{\psi}\gamma^5\psi. \tag{2.33}$$

Possible dimension 4 operators are

$$F_{\mu\nu}F^{\mu\nu}, \qquad F_{\mu\nu}\tilde{F}^{\mu\nu}, \qquad \bar{\psi}\gamma^\mu D_\mu\psi, \qquad \bar{\psi}\gamma^\mu D_\mu\gamma^5\psi, \tag{2.34}$$

where $\tilde{F}_{\mu\nu} \equiv (1/2)\epsilon_{\mu\nu\rho\sigma}F^{\rho\sigma}$.

By assuming canonically normalized kinetic term, we obtain the Lagrangian as

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m_R + im_I\gamma^5)\psi + \frac{e^2\theta}{16\pi^2}F_{\mu\nu}\tilde{F}^{\mu\nu}. \tag{2.35}$$

2.1.4 The phase in the mass and theta-term

In the standard textbook, m_I and θ are taken to be 0. Let us investigate m_I and θ in more detail. ψ bilinear term can be written as

$$m_R\bar{\psi}\psi - im_I\bar{\psi}\gamma^5\psi = (m_R + im_I)\bar{\psi}_R\psi_L + (m_R - im_I)\bar{\psi}_L\psi_R, \tag{2.36}$$

where $\psi_{L,R} = P_{L,R}\psi$. Thus, we can regard $m \equiv m_R + im_I$ as a complex mass parameter.

As similar to the discussion in the previous subsection, we replace ψ to ψ', and assume ψ' is related to ψ by the following chiral phase rotation:

$$\psi \to \psi' = \exp\left(\frac{i}{2}\xi\gamma_5\right)\psi, \tag{2.37}$$

where $\xi \equiv \arg(m_R + im_I)$. Then, we obtain

$$m_R\bar{\psi}'\psi' - im_I\bar{\psi}'\gamma^5\psi' = |m|\bar{\psi}\psi. \tag{2.38}$$

However, because of chiral anomaly, we obtain the following additional term in the Lagrangian:

$$\delta\mathcal{L} = \frac{\xi e^2}{16\pi^2} F_{\mu\nu}\tilde{F}^{\mu\nu}. \tag{2.39}$$

This effectively shift $\theta \to \theta + \xi$. Thus, we can conclude individual θ and $\arg m$ does not have physical meaning because they are basis-dependent. From these argument, we can see that the following two Lagrangians are equivalent:

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu D_\mu)\psi - m\bar{\psi}_R\psi_L - m^*\bar{\psi}_L\psi_R - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{\theta e^2}{16\pi^2}F_{\mu\nu}\tilde{F}^{\mu\nu}, \tag{2.40}$$

$$\mathcal{L}' = \bar{\psi}(i\gamma^\mu D_\mu)\psi - |m|\bar{\psi}\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{(\theta + \arg m)e^2}{16\pi^2}F_{\mu\nu}\tilde{F}^{\mu\nu}. \tag{2.41}$$

From this observation, we can define the following variable which is independent of the chiral phase rotation.

$$\bar{\theta} \equiv \theta + \arg m. \tag{2.42}$$

Does $\mathcal{L}_\theta$ have any physical effect? First $\mathcal{L}_\theta$ can be written as a total derivative as

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = \partial_\mu(\varepsilon^{\mu\nu\rho\sigma}A_\nu F_{\rho\sigma}). \tag{2.43}$$

This means there is no effect in perturbative calculation. Even in non-perturbative calculation, there is no configuration to contribute nonzero $\int d^4x F_{\mu\nu}\tilde{F}^{\mu\nu}$ in QED. Thus, there is no configuration in QED to make $\mathcal{L}_\theta$ relevant. (The existence of a monopole can be a loop hole of this argument. θ can have a physical effect in the presence of the monopoles [8, 9].)

To summarize, the most general renormalizable Lagrangian of QED automatically preserves C, P, and CP because $(e^2\theta/16\pi^2)F_{\mu\nu}\tilde{F}^{\mu\nu}$ term cannot have physical effect in QED. In this sense, C, P, CP are accidental symmetries in QED.

2.2 QCD

Now, let us talk about QCD, $SU(3)$ gauge theory with N_f flavors of fundamental representation quarks. The Lagrangian is

$$\mathcal{L} = -\frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu} + \frac{g^2\theta}{32\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu} + \sum_{i=1}^{N_f}\bar{q}_i(i\gamma^\mu D_\mu - m_i)q_i, \tag{2.44}$$

where the covariant derivative D_μ and field strength $G^a_{\mu\nu}$ are defined as

$$D_\mu = \partial_\mu + igA^a_\mu T^a, \tag{2.45}$$

$$G^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + gf^{abc}A^b_\mu A^c_\nu. \tag{2.46}$$

Here T^a is the generator of $SU(3)$ in the fundamental representation, and we take the normalization as $\mathrm{tr}[T^aT^b] = \delta^{ab}/2$. In this normalization, T^a is given by $T^a = \lambda^a/2$, where λ^a is the Gell-Mann matrix. It is convenient to define A_μ and $G_{\mu\nu}$ as

$$A_\mu \equiv A^a_\mu T^a, \qquad G_{\mu\nu} \equiv G^a_{\mu\nu}T^a = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]. \tag{2.47}$$

$$\mathcal{L} = -\frac{1}{2}\mathrm{tr}[G_{\mu\nu}G^{\mu\nu}] + \frac{g^2\theta}{16\pi^2}\mathrm{tr}[G_{\mu\nu}\tilde{G}^{\mu\nu}] + \sum_{i=1}^{N_f}\bar{q}_i(i\gamma^\mu D_\mu - m_i)q_i \tag{2.48}$$

The P transformation is defined as

$$q_i(t,\vec{x}) \to \gamma^0 q_i(t,-\vec{x}), \tag{2.49}$$
$$A^a_\mu(t,\vec{x}) \to P_\mu^{\ \nu} A^a_\nu(t,-\vec{x}). \tag{2.50}$$

The C transformation is defined as

$$q_i(t,\vec{x}) \to i\gamma^2 q_i^*(t,\vec{x}), \tag{2.51}$$
$$A^a_\mu(t,\vec{x})T^a \to -A^a_\mu(t,\vec{x})(T^a)^T. \tag{2.52}$$

Thus, each component of A^a_μ behaves as

$$A^{1,3,4,6,8}_\mu \to -A^{1,3,4,6,8}_\mu, \tag{2.53}$$
$$A^{2,5,7}_\mu \to A^{2,5,7}_\mu. \tag{2.54}$$

In the matrix form, the C transformation is simply written as

$$A_\mu(t,\vec{x}) \to -A^T_\mu(t,\vec{x}). \tag{2.55}$$

Let us show the θ term in QCD can be written as a total derivative.

$$\begin{aligned}
\mathrm{tr}[G_{\mu\nu}\tilde{G}^{\mu\nu}] &= \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}\mathrm{tr}[G_{\mu\nu}G_{\rho\sigma}] \\
&= \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}\mathrm{tr}[4(\partial_\mu A_\nu)(\partial_\rho A_\sigma) - 4ig(\partial_\mu A_\nu)[A_\rho, A_\sigma] - g^2[A_\mu, A_\nu][A_\rho, A_\sigma]] \\
&= \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}\mathrm{tr}[4(\partial_\mu A_\nu)(\partial_\rho A_\sigma) - 8ig(\partial_\mu A_\nu)A_\rho A_\sigma - g^2 A_\mu A_\nu[A_\rho, A_\sigma] + g^2 A_\mu[A_\rho, A_\sigma]A_\nu] \\
&= \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}\mathrm{tr}[4(\partial_\mu A_\nu)(\partial_\rho A_\sigma) - 8ig(\partial_\mu A_\nu)A_\rho A_\sigma] \\
&= \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}\partial_\mu\mathrm{tr}[4A_\nu(\partial_\rho A_\sigma) - \frac{8}{3}igA_\nu A_\rho A_\sigma].
\end{aligned} \tag{2.56}$$

Thus,

$$G^a_{\mu\nu}\tilde{G}^{a\mu\nu} = 2\mathrm{tr}[G_{\mu\nu}\tilde{G}^{\mu\nu}] = \partial_\mu J^\mu_{\mathrm{CS}}, \tag{2.57}$$

where J^μ_{CS} is the Chern–Simons current defined as

$$J^\mu_{\mathrm{CS}} \equiv 2\epsilon^{\mu\nu\rho\sigma}\mathrm{tr}[2A_\nu(\partial_\rho A_\sigma) - \frac{4}{3}igA_\nu A_\rho A_\sigma] = \epsilon^{\mu\nu\rho\sigma}\mathrm{tr}[2A_\nu G_{\rho\sigma} + \frac{4}{3}igA_\nu A_\rho A_\sigma]. \tag{2.58}$$

One of the most important differences between QED and QCD is that we *cannot* forget about θ term. In non-abelian gauge theory case, there is a configuration of gauge field to make θ term relevant, instanton. This will be discussed in the next section.

Under P and C transformation, $G_{\mu\nu}$ behaves as

$$G_{\mu\nu}(t,\vec{x}) \to_P P_\mu^{\ \alpha} P_\nu^{\ \beta} G_{\alpha\beta}(t,-\vec{x}), \qquad G_{\mu\nu}(t,\vec{x}) \to_C -G^T_{\mu\nu}(t,\vec{x}). \tag{2.59}$$

θ term can be written in the matrix form as

$$\mathcal{L}_\theta = \frac{\theta g^2}{16\pi^2}\mathrm{tr}[G_{\mu\nu}\tilde{G}^{\mu\nu}]. \tag{2.60}$$

Thus,

$$\mathcal{L}_\theta \to_P -\mathcal{L}_\theta, \qquad \mathcal{L}_\theta \to_C \mathcal{L}_\theta. \tag{2.61}$$

We can see that θ term preserves C but breaks both P and CP.

Let us write the complex mass term of quarks as

$$\mathcal{L} = -\frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu} + \sum_{i=1}^{N_f}[\bar{q}_i(i\gamma^\mu D_\mu)q_i - m_i\bar{q}_{i,R}q_{i,L} - m_i^*\bar{q}_{i,L}q_{i,R}]. \tag{2.62}$$

We can redefine the quark field by the following chiral phase rotation:

$$q_i \to q_i' = \exp\left(\frac{i}{2}\xi_i\gamma_5\right)q_i, \tag{2.63}$$

where $\xi_i = \arg(m_i)$. Then, we obtain

$$\mathcal{L}' = -\frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu} + \frac{g^2\sum_i\xi_i}{32\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu} + \sum_{i=1}^{N_f}[\bar{q}_i(i\gamma^\mu D_\mu)q_i - |m_i|\bar{q}_iq_i]. \tag{2.64}$$

From these argument, we can see that the following two Lagrangians are equivalent:

$$\mathcal{L} = -\frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu} + \frac{g^2\theta}{32\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu} + \sum_{i=1}^{N_f}[\bar{q}_i(i\gamma^\mu D_\mu)q_i - m_i\bar{q}_{i,R}q_{i,L} - m_i^*\bar{q}_{i,L}q_{i,R}], \tag{2.65}$$

$$\mathcal{L}' = -\frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu} + \frac{g^2(\theta + \sum_i\arg m_i)}{32\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu} + \sum_{i=1}^{N_f}[\bar{q}_i(i\gamma^\mu D_\mu)q_i - |m_i|\bar{q}_iq_i]. \tag{2.66}$$

We can define the following variable which is independent of the chiral phase rotation:

$$\bar{\theta} \equiv \theta + \sum_i \arg m_i. \tag{2.67}$$

2.3 Summary of this section

In this section, we wrote down the most generic renormalizable Lagrangian of QED and QCD. We found that θ term can be written in generic Lagrangian. However, θ term is total derivative, and we need more detailed discussion on its physical implications. As we will see in the next section, in non-abelian gauge theories, there is a non-perturbative contribution to make θ term relevant.

3 Instanton and vacua in non-abelian gauge theory

In this section, we discuss instanton and see that θ term in non-abelian gauge theory cannot be neglected in general. We also introduce θ-vacua. The following textbooks are useful to understand this section.

- Srednicki [10] §93
- Weinberg [11] §23
- Rubakov [12] §11, §13

3.1 Instanton

In the previous section, we have seen that θ term can be written as a total derivative. In this section, we show there exists a configuration such that the total derivative term can be relevant. This configuration is called instanton [13].

Let us discuss $SU(2)$ pure Yang–Mills theory in the Euclidean spacetime. The Lagrangian is given by

$$S_E = \int d^4x \frac{1}{4} F^a_{ij} F^{aij}. \tag{3.1}$$

By using the following identity,

$$\int d^4x_E (F^a_{ij} \pm \tilde{F}^a_{ij})^2 \geq 0, \tag{3.2}$$

we can obtain the following bound on the Euclidean action:

$$S_E \geq \frac{1}{4} \left| \int d^4x_E F^a_{ij} \tilde{F}^a_{ij} \right|. \tag{3.3}$$

Note that the integral in RHS can be written as a surface integral as

$$\begin{aligned} &\frac{g^2}{32\pi^2} \int d^4x_E F^a_{ij} \tilde{F}^{aij} \\ \to &\frac{g^2}{32\pi^2} \int d^4x_E \epsilon^{ijkl} \partial_i \left(\frac{4}{3} ig \mathrm{tr}[A_j A_k A_l] \right) \\ = &i \frac{g^3}{24\pi^2} \int dS_i \epsilon^{ijkl} \mathrm{tr}[A_j A_k A_l], \end{aligned} \tag{3.4}$$

where we have used F^a_{ij} decays fast enough at $|x| \to \infty$ and an asymptotic form of A_i at $|x| \to \infty$,

$$A_\mu \to \frac{i}{g} U \partial_\mu U^\dagger. \tag{3.5}$$

We obtain

$$n = \frac{1}{24\pi^2} \int dS_\mu \epsilon^{\mu\nu\rho\sigma} \mathrm{tr} \left[(U\partial_\nu U^\dagger)(U\partial_\rho U^\dagger)(U\partial_\sigma U^\dagger) \right]. \tag{3.6}$$

This n takes an integer value. Thus, the RHS of eq. (3.3) is discretized as

$$S_E \geq \frac{8\pi^2 |n|}{g^2}. \tag{3.7}$$

The lower bound of the Euclidean action is saturated when the following (anti-)self-dual condition is satisfied:

$$F^a_{ij} = \pm \tilde{F}^a_{ij}. \tag{3.8}$$

Let us take the following ansatz for the gauge field:

$$A_i(x) = \frac{i}{g} U_1 (\partial_i U_1^\dagger) f(r), \qquad U_1 = \frac{1}{r} \begin{pmatrix} x_4 + ix_3 & ix_1 + x_2 \\ ix_1 - x_2 & x_4 - ix_3 \end{pmatrix}, \tag{3.9}$$

where $r = \sqrt{x_1^2 + x_2^2 + x_3^2 + x_4^2}$. Note that this U_1 can be regarded as a nontrivial map from S^3 to $SU(2)$, and it has a winding number 1. By using this ansatz, we obtain

$$\partial_i A_j = \frac{1}{g} \left[-i(\partial_i \partial_j U_1) U_1^\dagger f + i(\partial_j U_1) U_1^\dagger (\partial_i U_1) U_1^\dagger f - i(\partial_j U_1) U_1^\dagger \partial_i f \right] \tag{3.10}$$

Then,

$$F_{ij} = \frac{1}{g} \left[\left[i(\partial_j U_1) U_1^\dagger (\partial_i U_1) U_1^\dagger (f - f^2) - i(\partial_j U_1) U_1^\dagger \partial_i f \right] - [i \leftrightarrow j] \right]. \tag{3.11}$$

At $(x_1, x_2, x_3, x_4) = (0, 0, 0, r)$, $U_1 = I_2$, $\partial_i U_1 = i\sigma^i / r$ $(i = 1, 2, 3)$ and $\partial_4 U_1 = 0$. Then,

$$gF_{12} = -\frac{i}{r^2} [i\sigma_1, i\sigma_2](f - f^2) = -\frac{2}{r^2}(f - f^2)\sigma^3, \tag{3.12}$$

$$gF_{34} = i\frac{i\sigma_3}{r} f' = -\frac{f'}{r}\sigma^3 \tag{3.13}$$

From $F_{12} = F_{34}$, we obtain

$$\frac{1}{r}\frac{df}{dr} = \frac{2}{r^2}(f - f^2). \tag{3.14}$$

This can be solved in a standard way:

$$\begin{aligned} \frac{df}{f - f^2} &= \frac{2}{r} dr \\ \log f - \log(1 - f) &= 2 \log \frac{r}{R}. \end{aligned} \tag{3.15}$$

where R is an arbitrary constant. Then, we easily obtain

$$\begin{aligned} \frac{1}{f} - 1 &= \frac{R^2}{r^2} \\ f &= \frac{r^2}{R^2 + r^2} \end{aligned} \tag{3.16}$$

To summarize, we obtain a non-trivial stationary point of Euclidean action:

$$A_i(x) = \frac{1}{g}\frac{r^2}{r^2+R^2} iU_1(\partial_i U_1^\dagger), \qquad U_1 = \frac{1}{r}\begin{pmatrix} x_4 + ix_3 & ix_1 + x_2 \\ ix_1 - x_2 & x_4 - ix_3 \end{pmatrix}. \tag{3.17}$$

From this configuration, we obtain

$$S_E = \frac{8\pi^2}{g^2}, \qquad \frac{g^2}{32\pi^2}\int d^4x F^a_{ij}\tilde{F}^{aij} = 1. \tag{3.18}$$

Note that the action of instanton is independent of size R of instanton. This is because Yang-Mills theory is classically scale-invariant.

3.2 Ground states in quantum pendulum

See Rubakov's textbook. [14, 15] would be also useful.

3.2.1 A ring on 3D space

The Lagrangian of a charged particle in vector potential $\vec{A}$ and potential $V(\vec{x})$ is given by

$$L = \frac{1}{2}m\dot{\vec{x}}^2 + e\vec{A}\cdot\dot{\vec{x}} - V(\vec{x}). \tag{3.19}$$

m is the mass of the particle and e is the charge of the particle. $V(\vec{x})$ can be gravitational potential, electric potential, or something else.

Let us assume this particle is constrained on a ring with radius ℓ. Then, we can write L as

$$L = \frac{1}{2}m\ell^2\dot{\varphi}^2 + eA_\varphi(\varphi)\ell\dot{\varphi} - V(\varphi). \tag{3.20}$$

A_φ is the φ-component of $\vec{A}$. By taking an appropriate gauge transformation, we can make A_φ to be a constant on the ring. Then, we can write A_φ as

$$A_\varphi = \frac{\Phi}{2\pi\ell}, \tag{3.21}$$

where Φ is the magnetic flux through the ring. Then, by defining $I \equiv m\ell^2$, we can write L as

$$L = \frac{1}{2}I\dot{\varphi}^2 + \frac{e\Phi}{2\pi}\dot{\varphi} - V(\varphi). \tag{3.22}$$

Then, we obtain the canonical momentum p_φ as

$$p_\varphi = \frac{\partial L}{\partial\dot{\varphi}} = I\dot{\varphi} + \frac{e\Phi}{2\pi}, \tag{3.23}$$

and the Hamiltonian H as

$$H = \frac{1}{2I}\left(p_\varphi - \frac{e\Phi}{2\pi}\right)^2 + V(\varphi). \tag{3.24}$$

Then Schrödinger equation is given by

$$\frac{1}{2I}\left(-i\frac{\partial}{\partial\varphi} - \frac{e\Phi}{2\pi}\right)^2\psi(\varphi) + V(\varphi)\psi(\varphi) = E\psi(\varphi), \tag{3.25}$$

where $\psi(\varphi)$ is the wave function with the periodic boundary condition

$$\psi(\varphi + 2\pi) = \psi(\varphi). \tag{3.26}$$

3.2.2 Generic Lagrangian of particle on S^1

Let us construct an equivalent setup from a different way.

First, let us assume $\varphi \in \mathbb{R}$. (not $[0, 2\pi)$!) By using a periodic potential $V(\varphi) = V(\varphi + 2\pi)$, we take the following Lagrangian:

$$L = \frac{1}{2} I \dot{\varphi}^2 - V(\varphi). \tag{3.27}$$

Then, we obtain the canonical momentum $p_\varphi = I\dot{\varphi}$ and the Hamiltonian H as

$$H = \frac{1}{2I} p_\varphi^2 + V(\varphi). \tag{3.28}$$

The Schrödinger equation is given by

$$-\frac{1}{2I} \frac{\partial^2}{\partial \varphi^2} \psi(\varphi) + V(\varphi)\psi(\varphi) = E\psi(\varphi). \tag{3.29}$$

This description has a redundancy because φ and $\varphi + 2\pi$ are physically equivalent. Thus, we need to impose a boundary condition on the wave function $\psi(\varphi)$:

$$|\psi(\varphi + 2\pi)| = |\psi(\varphi)|. \tag{3.30}$$

Then, generic boundary condition can be written as

$$\psi(\varphi + 2\pi) = e^{i\theta} \psi(\varphi), \tag{3.31}$$

where θ is a real parameter. Let us define

$$\tilde{\psi}(\varphi) \equiv \exp\left(-i \frac{\theta}{2\pi} \varphi\right) \psi(\varphi). \tag{3.32}$$

Then,

$$\tilde{\psi}(\varphi + 2\pi) = \tilde{\psi}(\varphi), \tag{3.33}$$

and the Schrödinger equation for $\tilde{\psi}$ can be written as

$$\frac{1}{2I} \left(-i \frac{\partial}{\partial \varphi} + \frac{\theta}{2\pi}\right)^2 \tilde{\psi}(\varphi) + V(\varphi)\tilde{\psi}(\varphi) = E\tilde{\psi}(\varphi). \tag{3.34}$$

Then, once we identify as

$$\theta = -e\Phi, \tag{3.35}$$

this setup is equivalent to the previous setup of a ring on 3D space!

3.2.3 n states and θ states on a ring

Let us assume $V(\varphi)$ is large enough to justify the WKB approximation. By assuming $V(\varphi)$ has a minimum at $\varphi = 0$, we can consider the wave function localized at $\varphi = 2\pi n$ such as

$$\psi_n(\varphi) \sim \exp\left(-\frac{1}{2}\frac{(\varphi - 2\pi n)^2}{\sigma^2}\right), \tag{3.36}$$

where $\sigma \ll 1$. Let us call this state as n states $|n\rangle$. However, n state does not satisfy the boundary condition (3.31). Thus, we can construct the following θ state which satisfies the boundary condition (3.31) by superposing n states:

$$\psi_\theta = \sum_{n=-\infty}^{\infty} e^{in\theta}\psi_n(\varphi). \tag{3.37}$$

This is the correct ground state of the system.

By assuming the tunneling between different n state is small, we can approximate the matrix element of the Hamiltonian H as

$$\langle m|H|n\rangle \simeq E_0\delta_{mn} - Ke^{-S}\delta_{m,n+1} - Ke^{-S}\delta_{m,n-1}. \tag{3.38}$$

Then, energy eigenstate $|\theta\rangle$ is given by

$$H|\theta\rangle = \left(E_0 - 2Ke^{-S}\cos\theta\right)|\theta\rangle. \tag{3.39}$$

3.3 Vacua in pure Yang–Mills theory

3.3.1 n vacua

We take temporal gauge $A_0 = 0$.

$$S = \int d^4x \left[\frac{1}{2}(\partial_0 A_i^a)^2 - \frac{1}{4}F_{ij}^a F^{aij}\right] \tag{3.40}$$

Let us discuss classical configuration to minimize the energy. Such a configuration should satisfy

$$F_{ij}^a = 0 \tag{3.41}$$

Then A_i can be written as a pure gauge configuration as

$$A_i = \frac{i}{g}U\partial_i U^\dagger. \tag{3.42}$$

Trivial vacuum field configuration $A_i^a = 0$ is obtained from $U(\vec{x}) = I_2$. However, apparently, there are other possible zero energy configurations!

There are field configurations with $U(\vec{x})$ which is separated from $A_i^a = 0$ by a potential barrier. However, we could have quantum tunneling between those field configurations. Such a tunneling is possible if

$$U(\vec{x}) \to I_2 \qquad \text{as } |\vec{x}| \to \infty. \tag{3.43}$$

Because of this boundary condition, $U(\vec{x})$ can be regarded as a map from S^3 to $SU(2)$. Such a map is classified by $\pi_3(SU(2)) = \mathbb{Z}$. Thus, we can label those field configurations by an integer n as

$$A_i^{(n)} = \frac{i}{g}U_n\partial_i U_n^\dagger. \tag{3.44}$$

	ring	pure Yang-Mills
Lagrangian	$(1/2)I\dot{\varphi}^2 - V(\varphi)$	$(1/4)F^a_{ij}F^{aij}$
Large gauge transf.	$\varphi \to \varphi + 2\pi$	$A_i \to UA_iU^\dagger + (i/g)U\partial_iU^\dagger$
n vacua	$\varphi \sim 2\pi n$	$A_i = (i/g)U\partial_iU^\dagger$
θ vacua	$\lvert\theta\rangle = \sum_n e^{in\theta}\lvert n\rangle$	$\lvert\theta\rangle = \sum_n e^{in\theta}\lvert n\rangle$
instanton	kink solution	BPST instanton

Table 2. Comparison between quantum pendulum and pure Yang-Mills theory.

3.3.2 θ vacua

These n vacua configurations are not continuously connected to each other. However, they are connected by large gauge transformation as

$$A_i^{(n)} = \frac{i}{g}U_n\partial_iU_n^\dagger = \frac{i}{g}U_1^n\partial_i(U_1^n)^\dagger. \tag{3.45}$$

Thus, the situation is quite similar to the quantum pendulum.

$$|\theta\rangle = \sum_{n=-\infty}^{\infty} e^{in\theta}|n\rangle. \tag{3.46}$$

3.4 Summary of this section

In this section, we have discussed the non-perturbative effect of non-abelian gauge theory, instanton. We also showed that instanton configuration can be regarded as a tunneling effect between n vacua, and the physical vacuum should be a linear combination of n vacua, θ vacua. Vacua of QCD also have similar structure as pure Yang-Mills theory. In this case, the physical parameter is $\bar{\theta} = \theta + \sum \arg m$. Thus, $\bar{\theta}$ cannot be physical if there exists a massless quark. If all quarks are massive, $\bar{\theta}$ is physical.

However, still, it is not clear what is the physical consequence of θ term. Since the θ term is a total derivative, we need to evaluate non-perturbative effects to understand its physical implications. For this purpose, in the next section, we will discuss physical consequences of θ term by using the chiral Lagrangian.

4 Chiral Lagrangian

In this section, we will discuss the physical consequence of $\bar{\theta}$ in QCD. Since θ term is a total derivative, it does not have any effect in perturbation theory. Thus, we need to understand non-perturbative effect of QCD to understand the physical consequence of θ term. For this purpose, we will use the effective theory for hadrons. The following textbooks and paper are useful to understand this section.

- Srednicki [10] §94
- Weinberg [11] §19
- Pich-de Rafael [16]

4.1 Global symmetry in QCD

Hadrons such as proton, neutron, pion, etc are understood as bound state of quarks and gluons, which can be described by QCD. Here we consider the QCD Lagrangian with three light quarks, up, down, and strange quarks. The Lagrangian is given by

$$\mathcal{L} = \sum_{q=u,d,s} [i\bar{q}\gamma^{\mu} D_{\mu} q - m_q \bar{q} q] \tag{4.1}$$

4.1.1 Chiral symmetry

In the massless limit, the Lagrangian is invariant under $U(3)_L \times U(3)_R$ chiral symmetry as

$$\begin{pmatrix} u_L \\ d_L \\ s_L \end{pmatrix} \to U_L \begin{pmatrix} u_L \\ d_L \\ s_L \end{pmatrix}, \qquad \begin{pmatrix} u_R \\ d_R \\ s_R \end{pmatrix} \to U_R \begin{pmatrix} u_R \\ d_R \\ s_R \end{pmatrix}, \tag{4.2}$$

where U_L and U_R are 3×3 unitary matrices. Thus, the kinetic term respects the following global symmetry:

$$U(3)_L \times U(3)_R \simeq SU(3)_L \times SU(3)_R \times U(1)_V \times U(1)_A. \tag{4.3}$$

It is known that the quark bilinear condensate is formed in the vacuum as

$$\langle \bar{u}u \rangle \simeq \langle \bar{d}d \rangle \simeq \langle \bar{s}s \rangle \simeq -\Lambda_{\mathrm{QCD}}^3. \tag{4.4}$$

Thus, the chiral symmetry is spontaneously broken, and the unbroken symmetry in the vacuum is a diagonal subgroup of $U(3)_L \times U(3)_R$ as

$$U(3)_V = SU(3)_V \times U(1)_V. \tag{4.5}$$

π, K, and η mesons are understood as NG bosons corresponding to the broken generators of $SU(3)_L \times SU(3)_R \to SU(3)_V$.

4.1.2 $U(1)_A$ violation in instanton background

For $U(1)_A$ symmetry, there is a source of explicit $U(1)_A$ violation in addition to spontaneous symmetry breaking by quark bilinear condensate.

First, because of chiral anomaly, the axial current J_A^{μ} is not conserved as [17, 18]

$$\partial_{\mu} J_A^{\mu} = -\frac{N_f g^2}{16\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}, \tag{4.6}$$

where N_f is the number of flavors. In this instanton background, $U(1)_A$ symmetry is explicitly broken by the chiral anomaly, and thus, we can have an extra $U(1)_A$ breaking effect in QCD.

4.2 $U(1)_A$ violation in chiral Lagrangian

The Lagrangian of QCD with three light quarks is

$$\mathcal{L} = \sum_{q=u,d,s} [i\bar{q}\gamma^\mu D_\mu q - m_q \bar{q}q] \tag{4.7}$$

In the limit of $m_u, m_d, m_s \to 0$, we have only the kinetic term of the quarks and the Lagrangian is invariant $U(3)_L \times U(3)_R$ chiral symmetry as eq. (4.2). Thus, there should be 9 massless NG bosons corresponding to generators of the broken symmetries.

Those spontaneously broken symmetries are actually *explicitly* broken by the quark mass terms. Since the light quark masses are smaller than Λ_{QCD}, those mass terms can be regarded as a perturbation. In this picture, we should have 9 NG bosons whose mass squared is proportional to the quark masses.

In this section, we will see that *this picture does not work*. We will see that there should be an *extra* source of $U(1)_A$ explicit breaking in addition to the quark masses [19]. We compare two EFTs with different symmetries:

- $SU(3)_L \times SU(3)_R \times U(1)_V \times U(1)_A$
- $SU(3)_L \times SU(3)_R \times U(1)_V$

We will see that the EFT with $U(1)_A$ fails to reproduce the observed spectrum of NG bosons, but the EFT without $U(1)_A$ does not have this problem.

4.2.1 EFT with $SU(3)_L \times SU(3)_R \times U(1)_V \times U(1)_A$

First, let us assume there is no extra $U(1)_A$ explicit breaking, and the spontaneous symmetry breaking pattern is given by $SU(3)_L \times SU(3)_R \times U(1)_V \times U(1)_A \to SU(3)_V \times U(1)_V$, and then we will see the mass spectrum of NG bosons from this assumption.

There are 9 broken generators, so we have 9 NG bosons. To write down the effective Lagrangian for NG bosons, we introduce a 3×3 unitary matrix Σ. Σ transforms under $U(3)_L \times U(3)_R$ as

$$\Sigma \to U_L \Sigma U_R^\dagger, \tag{4.8}$$

Then, $U(3)_L \times U(3)_R$ invariant Lagrangian for Σ is given by

$$\mathcal{L}_{\text{kin.}} = \frac{1}{4} f_\pi^2 \text{tr}[(\partial_\mu \Sigma)(\partial^\mu \Sigma^\dagger)] + \frac{1}{4} f'^2 (\partial_\mu \det \Sigma)(\partial^\mu \det \Sigma^\dagger), \tag{4.9}$$

where $f_\pi \simeq 93$ MeV is the pion decay constant.

$U(3)_L \times U(3)_R$ symmetry is explicitly broken by quark mass term. The quark mass matrix $M = \text{diag}(m_u, m_d, m_s)$ can be regarded as a spurion field which transforms under $U(3)_L \times U(3)_R$ as

$$M \to U_R M U_L^\dagger. \tag{4.10}$$

Then, the leading order term of M in the effective Lagrangian is given by

$$\mathcal{L}_m = -V = \frac{1}{2} f_\pi^2 B_0 \mathrm{tr}[M\Sigma + M^\dagger \Sigma^\dagger] \tag{4.11}$$

Σ can be written in terms of NG bosons as

$$\Sigma = \exp\left(\frac{i}{f_\pi}\Phi + \frac{i}{f_{\eta'}}\sqrt{\frac{2}{3}}\eta' \right) \tag{4.12}$$

where $f_{\eta'} \equiv \sqrt{f_\pi^2 + 3f'^2}$ and

$$\Phi = \begin{pmatrix} \pi_0 + \eta/\sqrt{3} & \sqrt{2}\pi_+ & \sqrt{2}K_+ \\ \sqrt{2}\pi_- & -\pi^0 + \eta/\sqrt{3} & \sqrt{2}K_0 \\ \sqrt{2}K_- & \sqrt{2}\bar{K_0} & -2\eta/\sqrt{3} \end{pmatrix}. \tag{4.13}$$

V can be expanded as

$$\begin{aligned} V =& -f_\pi^2 B_0 (m_u + m_d + m_s) \\ & + (m_u + m_d) B_0 \pi_+ \pi_- + (m_u + m_s) B_0 K_+ K_- + (m_d + m_s) B_0 K_0 \bar{K}_0 + \frac{1}{2} v_0^T M_0^2 v_0 \\ & + \cdots \end{aligned} \tag{4.14}$$

where v_0 and M_0^2 are defined as

$$v_0 = \begin{pmatrix} \pi_0 \\ \eta \\ \eta' \end{pmatrix}, \tag{4.15}$$

$$M_0^2 = B_0 \begin{pmatrix} m_u + m_d & \frac{1}{\sqrt{3}}(m_u - m_d) & \sqrt{\frac{2}{3}}\frac{f_\pi}{f_{\eta'}}(m_u - m_d) \\ \frac{1}{\sqrt{3}}(m_u - m_d) & \frac{1}{3}(m_u + m_d + 4m_s) & \frac{\sqrt{2}}{3}\frac{f_\pi}{f_{\eta'}}(m_u + m_d - 2m_s) \\ \sqrt{\frac{2}{3}}\frac{f_\pi}{f_{\eta'}}(m_u - m_d) & \frac{\sqrt{2}}{3}\frac{f_\pi}{f_{\eta'}}(m_u + m_d - 2m_s) & \frac{2}{3}\frac{f_\pi^2}{{f_{\eta'}}^2}(m_u + m_d + m_s) \end{pmatrix}. \tag{4.16}$$

Let us discuss the mass spectrum in the limit of $m_u \sim m_d \ll m_s$. In this limit,

$$M_0^2 = m_s B_0 \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{4}{3} & -\frac{2\sqrt{2}}{3}\frac{f_\pi}{f_{\eta'}} \\ 0 & -\frac{2\sqrt{2}}{3}\frac{f_\pi}{f_{\eta'}} & \frac{2}{3}\frac{f_\pi^2}{{f_{\eta'}}^2} \end{pmatrix} + \mathcal{O}(m_{u,d}B_0). \tag{4.17}$$

The mass eigenstates at the leading order are described as the following unit vectors:

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \tag{4.18}$$

$$v_2 = \frac{1}{\sqrt{f_\pi^2 + 2{f_{\eta'}}^2}} \begin{pmatrix} 0 \\ f_\pi \\ \sqrt{2}f_{\eta'} \end{pmatrix}, \tag{4.19}$$

$$v_3 = \frac{1}{\sqrt{f_\pi^2 + 2{f_{\eta'}}^2}} \begin{pmatrix} 0 \\ -\sqrt{2}f_{\eta'} \\ f_\pi \end{pmatrix}. \tag{4.20}$$

The mass eigenvalue for v_1 and v_2 are $\mathcal{O}(m_{u,d}B_0)$, and thus, they are light states. However, the mass eigenvalue for v_3 is $\mathcal{O}(m_s B_0)$, and thus, it is a heavy state. The mass eigenvalue for v_3 is given by

$$m_3^2 = \left(\frac{4}{3} + \frac{2}{3}\frac{f_\pi^2}{{f_{\eta'}}^2}\right) m_s B_0 + \mathcal{O}(m_{u,d}B_0). \tag{4.21}$$

To discuss the mass eigenvalue for v_1 and v_2, let us define $\tilde{M}_0^2$ as

$$\tilde{M}_0^2 = \begin{pmatrix} m_{11}^2 & m_{12}^2 \\ m_{21}^2 & m_{22}^2 \end{pmatrix}, \tag{4.22}$$

where

$$m_{11}^2 = v_1^T M_0^2 v_1 = (m_u + m_d)B_0, \tag{4.23}$$

$$m_{12}^2 = m_{21}^2 = v_1^T M_0^2 v_2 = \frac{\sqrt{3}(m_u - m_d)B_0}{\sqrt{1 + 2(f_{\eta'}^2/f_\pi^2)}}, \tag{4.24}$$

$$m_{22}^2 = v_2^T M_0^2 v_2 = \frac{3(m_u + m_d)B_0}{1 + 2(f_{\eta'}^2/f_\pi^2)}. \tag{4.25}$$

By neglecting m_{12}^2, m_{11}^2 corresponds to m_π^2, and m_{22}^2 is also a mass eigenvalue as

$$m_{22}^2 \simeq \frac{3}{1 + 2(f_{\eta'}^2/f_\pi^2)}(m_u + m_d)B_0 \leq 3m_\pi^2 \sim (240\text{ MeV})^2. \tag{4.26}$$

However, we know $m_\eta \simeq 548$ MeV and $m_{\eta'} \simeq 958$ MeV. There is no such light state as m_2 in the observed spectrum!

4.2.2 EFT with $SU(3)_L \times SU(3)_R \times U(1)_V$: an extra $U(1)_A$ breaking

We have seen that the EFT with $SU(3)_L \times SU(3)_R \times U(1)_V \times U(1)_A$ predicts a light pseudoscalar meson which is not observed in the real world.

Let us add an extra $U(1)_A$ breaking term to the effective Lagrangian by hand as

$$\mathcal{L}_{\text{break}} = -\frac{1}{2}M^2\eta'^2 = -\frac{1}{12}f_{\eta'}^2 M^2(\arg\det\Sigma)^2. \tag{4.27}$$

This term explicitly breaks $U(1)_A$ symmetry but preserves $SU(3)_L \times SU(3)_R \times U(1)_V$ symmetry.

By assuming $M^2 \gg m_s B_0$, we can integrate out η'. We can simply write Σ as

$$\Sigma = \exp\left(\frac{i}{f_\pi}\Phi\right), \tag{4.28}$$

and then, V in eq. (4.11) can be expanded as

$$\begin{aligned} V =& - f_\pi^2 B_0(m_u + m_d + m_s) \\ &+ (m_u + m_d)B_0\pi_+\pi_- + (m_u + m_s)B_0K_+K_- + (m_d + m_s)B_0K_0\bar{K}_0 + \frac{1}{2}B_0 v_0^T M_0^2 v_0 \\ &+ \cdots , \end{aligned} \tag{4.29}$$

where v_0 and M_0^2 are defined as

$$v_0 = \begin{pmatrix} \pi_0 \\ \eta \end{pmatrix}, \tag{4.30}$$

$$M_0^2 = \begin{pmatrix} m_u + m_d & \frac{1}{\sqrt{3}}(m_u - m_d) \\ \frac{1}{\sqrt{3}}(m_u - m_d) & \frac{1}{3}(m_u + m_d + 4m_s) \end{pmatrix}. \tag{4.31}$$

In the limit of $m_u \sim m_d \ll m_s$, the mass eigenstates are approximately given by π_0 and η. Then, the mass of NG bosons are given by

$$m_{\pi_0}^2 = m_{\pi_\pm}^2 = (m_u + m_d)B_0, \tag{4.32}$$

$$m_{K_\pm}^2 = (m_u + m_s)B_0, \tag{4.33}$$

$$m_{K_0}^2 = (m_d + m_s)B_0, \tag{4.34}$$

$$m_\eta^2 = \frac{1}{3}(m_u + m_d + 4m_s)B_0 \tag{4.35}$$

We can see that this mass spectrum of NG bosons satisfy the following relation:

$$3m_\eta^2 + m_\pi^2 = 2m_{K_\pm}^2 + 2m_{K_0}^2 \tag{4.36}$$

This is so-called Gell-Mann–Okubo formula [20–22]. The observed values of NG boson masses are $m_{\pi_\pm} \simeq 140$ MeV, $m_{\pi_0} \simeq 135$ MeV, $m_\eta \simeq 548$ MeV, $m_{K_\pm} \simeq 494$ MeV, and $m_{K_0} \simeq 498$ MeV. Then, we obtain

$$\frac{3m_\eta^2 + m_\pi^2}{2m_{K_\pm}^2 + 2m_{K_0}^2} \simeq 0.93. \tag{4.37}$$

Thus, the Gell-Mann–Okubo formula is satisfied with good accuracy. This fact strongly supports the validity of the effective theory for NG bosons.

4.3 Strong CP violation in chiral Lagrangian

As we have learned so far, the QCD Lagrangian in the chiral limit ($m = 0$) has $U(1)_A$ symmetry at the classical level. However, this $U(1)_A$ symmetry is explicitly broken by the non-perturbative effect of QCD. For example, in the instanton background, $U(1)_A$ symmetry is explicitly broken by the chiral anomaly. This $U(1)_A$ breaking effect is important to reproduce the observed spectrum of NG bosons.

In this section, we will see that, once we accept the existence of this $U(1)_A$ explicit breaking effect, a CP violating parameter in QCD becomes physical and the strong CP problem arises if there is no massless quark.

4.3.1 EFT with CP violation

Now let us discuss the effect of nonzero $\bar{\theta}$. See e.g. ref. [16, 23]. It is convenient to eliminate θ term by chiral rotation of quark fields. Since $\bar{\theta}$ is invariant under such a redefinition, we obtain the following Lagrangian:

$$\mathcal{L} = -\frac{1}{4}G_{\mu\nu}^a G^{a\mu\nu} + \sum_q \left[i\bar{q}\gamma^\mu D_\mu q - (\bar{q}_R m_q e^{i\theta_q} q_L + h.c.) \right], \tag{4.38}$$

where

$$\bar{\theta} = \theta_u + \theta_d + \theta_s. \tag{4.39}$$

Let us discuss the VEV of Σ. By assuming the form of Σ as

$$\Sigma = \mathrm{diag}(e^{i\varphi_u}, e^{i\varphi_d}, e^{i\varphi_s}), \tag{4.40}$$

where $\varphi_u + \varphi_d + \varphi_s = 0$. We obtain the following potential for φ_q:

$$V = -f_\pi^2 B_0 \sum_q m_q \cos(\theta_q + \varphi_q). \tag{4.41}$$

Let us minimize V with respect to φ_q under the constraint $\sum_q \varphi_q = 0$ by using Lagrange multiplier λ. [24, 25]

$$\frac{\partial V}{\partial \varphi_q} - \lambda = f_\pi^2 B_0 m_q \sin(\theta_q + \varphi_q) - \lambda = 0. \tag{4.42}$$

This is the Dashen-Nuyts equation. By assuming $\varphi_q,\ \theta_q \ll 1$, we obtain

$$\varphi_q = -\theta_q + \frac{\lambda}{f_\pi^2 B_0 m_q}. \tag{4.43}$$

From $\sum_q \varphi_q = 0$, we obtain

$$\lambda = f_\pi^2 B_0 m_* \bar{\theta}, \tag{4.44}$$

where m_* is defined as

$$m_* = \left(\frac{1}{m_u} + \frac{1}{m_d} + \frac{1}{m_s} \right)^{-1} \tag{4.45}$$

Thus, φ_q is given by

$$\varphi_q = -\theta_q + \frac{m_*}{m_q}\bar{\theta}. \tag{4.46}$$

We can see that $\langle \Sigma \rangle \neq I$ in general.

It is useful to take a basis in which $\varphi_q = 0$ so that $\langle \Sigma \rangle = I$. In this basis, θ_q is given by

$$\theta_u = \frac{m_*}{m_u}\bar{\theta}, \qquad \theta_d = \frac{m_*}{m_d}\bar{\theta}, \qquad \theta_s = \frac{m_*}{m_s}\bar{\theta}, \tag{4.47}$$

Note that, in this basis,

$$m_u \theta_u = m_d \theta_d = m_s \theta_s. \tag{4.48}$$

To summarize, for nonzero $\bar{\theta}$, the following chiral Lagrangian is convenient:

$$\mathcal{L} = \frac{1}{4} f_\pi^2 \mathrm{tr}[(\partial_\mu \Sigma)(\partial^\mu \Sigma^\dagger)] + \frac{1}{2} f_\pi^2 B_0 \mathrm{tr}[M\Sigma + M^\dagger \Sigma^\dagger], \tag{4.49}$$

$$M = \mathrm{diag}\big(m_u \exp\big(i m_* \bar{\theta}/m_u\big), m_d \exp\big(i m_* \bar{\theta}/m_d\big), m_s \exp\big(i m_* \bar{\theta}/m_s\big)\big). \tag{4.50}$$

For $\bar{\theta} \ll 1$, we obtain

$$M \simeq \mathrm{diag}(m_u, m_d, m_s) + i m_* \bar{\theta} I, \tag{4.51}$$

and $\langle \Sigma \rangle = I$ is satisfied.

4.3.2 $\eta \to \pi\pi$

Let us expand the quark mass dependent term in the effective Lagrangian. $\Sigma = \exp(i\Phi/f_\pi)$, $M \to M + im_*\bar{\theta}I$. Then, we obtain [16, 26, 27]

$$\begin{aligned}\mathcal{L} &\ni \frac{1}{2}f_\pi^2 B_0 \mathrm{tr}[(M + im_*\bar{\theta}I)\Sigma + (M - im_*\bar{\theta}I)\Sigma^\dagger] \\ &\simeq f_\pi^2 B_0 \mathrm{tr} M - \frac{1}{2}B_0 \mathrm{tr}[M\Phi^2] + \frac{m_*\bar{\theta}B_0}{6f_\pi}\mathrm{tr}[\Phi^3] + \cdots . \end{aligned} \tag{4.52}$$

Thus, we obtained CP violating cubic interaction term. In particular, there are the following interaction terms that contribute to the CP violating decay η decays:

$$\mathcal{L} \ni \frac{2\bar{\theta}m_* B_0}{\sqrt{3}f_\pi}\left(\eta\pi_+\pi_- + \frac{1}{2}\eta\pi_0^2\right) \tag{4.53}$$

This induces the CP violating decay $\eta \to \pi_+\pi_-$, which is constrained as $\mathrm{Br}(\eta \to \pi_+\pi_-) < 4.4 \times 10^{-6}$ [28]. By using the chiral Lagrangian, we can estimate the branching fraction of $\eta \to \pi_+\pi_-$ as [27]

$$\mathrm{Br}(\eta \to \pi_+\pi_-) \sim 10^2 \times \bar{\theta}^2. \tag{4.54}$$

From this measurement, we obtain $|\bar{\theta}| \lesssim 10^{-4}$.

4.4 Neutron EDM and chiral Lagrangian with baryon

The most important consequence of nonzero $\bar{\theta}$ is the neutron EDM. Here we calculate the neutron EDM induced from nonzero $\bar{\theta}$ [16, 23]. For details, see §83 and §94 in Srednicki's book and "Weak Interactions and Modern Particle Theory" by Georgi.

Let us discuss chiral Lagrangian with two flavors of quarks. The nucleon doublet N, and the meson field Σ are defined as

$$N = \begin{pmatrix} p \\ n \end{pmatrix}, \qquad \Sigma = \exp\left(\frac{i}{f_\pi}\Phi\right), \qquad \Phi = \begin{pmatrix} \pi^0 & \sqrt{2}\pi^+ \\ \sqrt{2}\pi^- & -\pi^0 \end{pmatrix}. \tag{4.55}$$

We assume left-handed and right-handed nucleon fields $N_L \equiv P_L N$ and $N_R \equiv P_R N$ is transformed under $SU(2)_L \times SU(2)_R$ as[2]

$$N_L \to g_L N_L, \qquad N_R \to g_R N_R. \tag{4.57}$$

Then, by requiring the invariance under parity transformation,

$$N_L \leftrightarrow N_R, \qquad \Sigma \leftrightarrow \Sigma^\dagger, \qquad M \leftrightarrow M^\dagger, \tag{4.58}$$

[2]Note that the nucleon field in CCWZ prescription [29, 30] is transformed as $N \to hN$, and the NG boson fields ξ_L and ξ_R are transformed as $\xi_L \to g_L \xi_L h^\dagger$ and $\xi_R \to g_R \xi_R h^\dagger$. The meson field Σ and the nucleon field N defined in eq. (4.57) is written as

$$N = (\xi_L P_L N + \xi_R P_R N)|_{\mathrm{CCWZ}}, \qquad \Sigma = \xi_L \xi_R^\dagger|_{\mathrm{CCWZ}}. \tag{4.56}$$

we obtain the quark mass M independent term as

$$\begin{aligned}\mathcal{L}_0 =& i\bar{N}\gamma^\mu\partial_\mu N - m_N(\bar{N}_R\Sigma^\dagger N_L + \bar{N}_L\Sigma N_R)\\ &- \frac{1}{2}(g_A-1)i(\bar{N}_L\gamma^\mu\Sigma(\partial_\mu\Sigma^\dagger)N_L + \bar{N}_R\gamma^\mu\Sigma^\dagger(\partial_\mu\Sigma)N_R)\end{aligned} \tag{4.59}$$

$\mathcal{O}(M^1)$ terms are given by

$$\begin{aligned}\mathcal{L}_1 =& - c_1(\bar{N}_R M^\dagger N_L + \bar{N}_L M N_R) - c_2(\bar{N}_R\Sigma^\dagger M\Sigma^\dagger N_L + \bar{N}_L\Sigma M^\dagger\Sigma N_R)\\ &- c_3\mathrm{tr}[M\Sigma^\dagger + M^\dagger\Sigma](\bar{N}_R\Sigma^\dagger N_L + \bar{N}_L\Sigma N_R)\\ &- c_4\mathrm{tr}[M\Sigma^\dagger - M^\dagger\Sigma](\bar{N}_R\Sigma^\dagger N_L - \bar{N}_L\Sigma N_R).\end{aligned} \tag{4.60}$$

It is convenient to redefine N_L and N_R as

$$N_L = \Sigma^{1/2}\mathcal{N}_L, \qquad N_R = \Sigma^{-1/2}\mathcal{N}_R. \tag{4.61}$$

Then, eq. (4.59) can be rewritten as

$$\mathcal{L}_0 = i\bar{\mathcal{N}}\gamma^\mu\partial_\mu\mathcal{N} - m_N\bar{\mathcal{N}}\mathcal{N} + \frac{g_A}{2f_\pi}\bar{\mathcal{N}}\gamma^\mu\gamma^5(\partial_\mu\Phi)\mathcal{N} + \cdots \tag{4.62}$$

Let us expand M as

$$M \to \hat{M} + im_*\bar{\theta}I, \tag{4.63}$$

where $m_* = (1/m_u + 1/m_d)^{-1}$ and $\hat{M} = \mathrm{diag}(m_u, m_d)$. Then, we obtain $\bar{\theta}$-dependent terms in eq. (4.60) as[3]

$$\mathcal{L}_1 \ni -\frac{m_*\bar{\theta}}{f_\pi}(c_1+c_2)\bar{\mathcal{N}}\Phi\mathcal{N} + im_*\bar{\theta}(-c_- + 4c_4)\bar{\mathcal{N}}\gamma_5\mathcal{N}, \tag{4.64}$$

where $c_\pm = c_1 \pm c_2$. Thus, the relevant interaction terms for neutron EDM in eqs. (4.62) and (4.64) are given by

$$\mathcal{L} \ni \frac{g_A}{\sqrt{2}f_\pi}((\partial_\mu\pi^-)\bar{n}\gamma^\mu\gamma^5 p + (\partial_\mu\pi^+)\bar{p}\gamma^\mu\gamma^5 n) - \frac{\sqrt{2}m_*\bar{\theta}c_+}{f_\pi}(\pi^-\bar{n}p + \pi^+\bar{p}n) + \cdots \tag{4.65}$$

c_+ is determined by comparing three-flavor chiral Lagrangian with the observed baryon masses as

$$c_+ = \frac{m_\Xi - m_\Sigma}{2(m_K^2 - m_\pi^2)}\frac{m_\pi^2}{\bar{m}} \simeq 1.7. \tag{4.66}$$

d_n can be evaluated by the one-loop diagram with charged pion loop. In particular, the chiral log contribution is given by

$$d_n \sim \frac{e}{16\pi^2}\frac{m_*\bar{\theta}c_+}{f_\pi}\frac{g_A m_N}{f_\pi}\frac{1}{m_N}\log\frac{\Lambda}{m_\pi} \sim e\bar{\theta}\times\frac{c_+ g_A m_*}{(4\pi f_\pi)^2}\log\frac{m_N}{m_\pi} \sim \bar{\theta}\times 10^{-16}\ e\ \mathrm{cm}, \tag{4.67}$$

where $\Lambda \sim 1$ GeV. Note that d_n contains an unknown counterterm. So we should understand that this estimation has $\mathcal{O}(1)$ uncertainty.

Ref.	
Dragos et al (2019) [31]	$d_n = \bar{\theta} \times (-1.52 \pm 0.71) \times 10^{-16}$ e cm
Alexandrou et al (2020) [32]	$\lvert d_n \rvert = \bar{\theta} \times (-0.9 \pm 2.4) \times 10^{-16}$ e cm
Bhattacharya et al (2021) [33]	$d_n = \bar{\theta} \times (-3 \pm 7 \pm 20) \times 10^{-16}$ e cm
χQCD (2023) [34]	$d_n = \bar{\theta} \times (-1.48 \pm 0.14 \pm 0.31) \times 10^{-16}$ e cm
Hisano et al (2012) [35]	$d_n = \bar{\theta} \times 0.4 \times 10^{-16}$ e cm
Ema et al (2024) [36]	$d_n = \bar{\theta} \times (0.5 - 1.5) \times 10^{-16}$ e cm

Table 3. Lattice QCD and QCD sum rule results for neutron EDM induced by $\bar{\theta}$ term.

4.5 More on neutron EDM

So far we have estimated the neutron EDM induced by $\bar{\theta}$ term by using the chiral Lagrangian. We have seen that one-loop diagram with charged pion loop gives the neutron EDM. However, this calculation suffers from the uncertainty of counter term and it is not possible to predict with $\mathcal{O}(1)$ accuracy.

In table 3, we summarize the results of lattice QCD calculations and QCD sum rules for neutron EDM induced by $\bar{\theta}$ term. We can see that the results still have $\mathcal{O}(1)$ uncertainty, but they are more or less consistent with the estimation by chiral Lagrangian, and

$$d_n \sim \bar{\theta} \times 10^{-16} \ e \ \text{cm}. \tag{4.68}$$

4.6 Summary of this section

In this section, we analyzed the chiral Lagrangian to discuss physics of hadrons. We found that

- In addition to the quark mass terms, we need to add a $U(1)_A$ breaking term. Then, we can reproduce the correct spectrum of NG bosons, which is consistent with the Gell-Mann–Okubo formula.

- If we do not add $U(1)_A$ breaking term, we predict a light pseudoscalar meson whose mass is smaller than ~ 240 MeV in addition to π^0. Such a meson is not observed in the real world.

- Once we accept the existence of $U(1)_A$ breaking effect, a CP violating parameter $\bar{\theta}$ inevitably becomes physical and induces CP violating processes such as $\eta \to \pi\pi$ and neutron EDM.

In the next section, we will discuss CP violation in the Standard Model. Actually, the situation becomes worse in the Standard Model because there is another source of CP violation, the CP violating phase in the CKM matrix. We will see that the smallness of $\bar{\theta}$ is quite unnatural in the Standard Model.

[3] We also obtain complex baryon mass term $\sim i\bar{\theta}\bar{\mathcal{N}}\gamma^5\mathcal{N}$ but this gives subdominant contribution to neutron EDM.

	$SU(3)_C$	$SU(2)_L$	$U(1)_Y$
$q_{L,i} = (u_{L,i}, d_{L,i})$	3	2	1/6
$u_{R,i}$	3	1	2/3
$d_{R,i}$	3	1	$-1/3$
$\ell_{L,i}$	1	2	$-1/2$
$e_{R,i}$	1	1	-1
H	1	2	1/2

Table 4. Matter content of the Standard Model. $i = 1, 2, 3$ express the generation index.

5 CP violation in the Standard Model

Let us discuss CP violation in the Standard Model. As we have seen that there is the strong CP problem in QCD, i.e., we could write CP violating term in the Lagrangian but somehow the nature respects CP symmetry. Actually, the situation becomes much worse in the Standard Model. This is because there is another source of CP violation in the Standard Model, and indeed, we know that its effect is sizable! In this sense, the smallness of $\bar{\theta}$ in QCD is quite unnatural. We need to have a hierarchy between two CP violating parameters, $\bar{\theta}$ and the CP violating phase in the CKM matrix. However, there is no reason for such a hierarchy in the Standard Model. In this section, we will discuss the strong CP problem in the Standard Model. The following textbooks are useful to understand this section.

- Peskin-Schroeder [7] §20
- Srednicki [10] §87, §88, §89

5.1 The number of parameters in the Standard Model

The Standard Model (SM) is a $SU(3)_C \times SU(2)_L \times U(1)_Y$ gauge theory with the following matter content shown in table 4. The SM has three generations of quarks and leptons.

Let us write down the most generic renormalizable Lagrangian of the Standard Model as we did in the QED and QCD cases. The Lagrangian of the Standard Model is given by

$$\mathcal{L} = \mathcal{L}_{\text{kin}} + \mathcal{L}_{\text{yuk}} + \mathcal{L}_H + \mathcal{L}_\theta, \tag{5.1}$$

where $\mathcal{L}_{\text{kin}}$ is the kinetic term of gauge bosons, quarks, leptons, and Higgs field. $\mathcal{L}_{\text{yuk}}$, $\mathcal{L}_H$, and $\mathcal{L}_\theta$ are given by

$$\mathcal{L}_{\text{yuk}} = -y_{u,ij}\bar{u}_{R,i}\tilde{H}^\dagger q_{L,j} - y_{d,ij}\bar{d}_{R,i}H^\dagger q_{L,j} - y_{e,ij}\bar{e}_{R,i}H^\dagger \ell_{L,j} + \text{h.c.}, \tag{5.2}$$

$$\mathcal{L}_H = -\lambda|H|^4 + \mu^2|H|^2, \tag{5.3}$$

$$\mathcal{L}_\theta = \frac{\theta_Y g'^2}{16\pi^2}B_{\mu\nu}\tilde{B}^{\mu\nu} + \frac{\theta_2 g_2^2}{32\pi^2}W^a_{\mu\nu}\tilde{W}^{a\mu\nu} + \frac{\theta g_s^2}{32\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu}. \tag{5.4}$$

where H is a $SU(2)_L$ doublet Higgs field with $Y = +1/2$ and $\tilde{H} = i\sigma^2 H^*$.

How many parameters does the Standard Model have? The Yukawa matrices can be rewritten by singular value decomposition as

$$y_u = U_{u_R} \hat{y}_u U_{u_L}^\dagger, \qquad y_d = U_{d_R} \hat{y}_d U_{d_L}^\dagger, \qquad y_e = U_{e_R} \hat{y}_e U_{e_L}^\dagger, \tag{5.5}$$

We can define the Cabibbo–Kobayashi–Maskawa (CKM) matrix as

$$V_{\text{CKM}} = U_{u_L}^\dagger U_{d_L}. \tag{5.6}$$

3×3 unitary matrix has 9 parameters, but we can eliminate 5 parameters by rephasing of quark fields. Thus, the CKM matrix has 4 physical parameters.

Let us discuss the θ term in the Standard Model. First θ_Y cannot have any physical effect as we have discussed in the QED case. How about θ_2? Let us consider the phase rotation by lepton number as

$$\ell_{L,i} \to e^{i\alpha} \ell_{L,i}, \qquad e_{R,i} \to e^{i\alpha} e_{R,i}. \tag{5.7}$$

$\mathcal{L}_{\text{Yuk}}$ and $\mathcal{L}_H$ are invariant under this phase rotation, but $\mathcal{L}_\theta$ is not invariant because of chiral anomaly. By using Fujikawa method, we can find the following shift of θ parameters:

$$\theta_Y \to \theta_Y + \frac{1}{2} N_g \alpha, \qquad \theta_2 \to \theta_2 - N_g \alpha, \tag{5.8}$$

where $N_g = 3$ is the number of generations. Thus, θ_2 can be eliminated by the lepton number phase rotation. Thus, both θ_Y and θ_2 do not have any physical effect. θ for gluon can be shifted by a chiral rotation of quark field such as $u_R \to e^{i\alpha} u_R$. However, this necessarily shift the phase of the determinant of the Yukawa matrices. Then, we can find that $\bar{\theta}$ defined below is invariant under the chiral rotation, namely, $\bar{\theta}$ is a physical parameter.

$$\bar{\theta} = \theta + \arg\det y_u y_d. \tag{5.9}$$

On the other hand, $\bar{\theta}$ cannot be eliminated. Thus, $\bar{\theta}$ is a physical parameter in the Standard Model.

To summarize, the Standard Model has 19 physical parameters.

- 3 gauge coupling constants (g_1, g_2, g_3)
- 2 Higgs potential parameters (λ, μ^2)
- 9 fermion masses (m_u, m_d, m_s, m_c, m_b, m_t, m_e, m_μ, m_τ)
- 4 parameter in CKM matrix
- 1 $\bar{\theta}$ parameter in QCD

5.2 CP violation in the Standard Model

In the SM, P is maximally violated because only left-handed quarks and leptons have $SU(2)_L$ gauge interaction.

Let us discuss CP symmetry. We define CP transformation as

$$q_i(t,\vec{x}) \to i\gamma^0\gamma^2 q_i^*(t,-\vec{x}). \tag{5.10}$$

We can see that $\mathcal{L}$ is invariant under CP if y_u, y_d, and y_e are real and $\theta = 0$. This corresponds to real V_{CKM} and $\bar{\theta} = 0$.

5.2.1 CP violation in the CKM matrix

CKM matrix can be written as

$$V_{\mathrm{CKM}} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}. \tag{5.11}$$

Under $q \to q' = e^{i\alpha}q$, the CKM matrix is transformed as

$$V_{\mathrm{CKM}} \to \mathrm{diag}(e^{i\alpha_u}, e^{i\alpha_c}, e^{i\alpha_t}) V_{\mathrm{CKM}} \mathrm{diag}(e^{-i\alpha_d}, e^{-i\alpha_s}, e^{-i\alpha_b}). \tag{5.12}$$

Physical parameter should be invariant under this transformation. Thus, CKM matrix has 4 physical parameters. If CKM matrix were real, CKM matrix is real orthogonal matrix, and it can be parametrized by three mixing angles. Thus, the remaining one parameter is a CP violating phase δ_{CKM}.

The components of V_{CKM} has the following hierarchical structure:

- $|V_{ud}|, |V_{cs}|, |V_{tb}| \sim 1$
- $|V_{us}|, |V_{cd}| \sim 0.22$
- $|V_{cb}|, |V_{ts}| \sim 0.04$
- $|V_{ub}| \sim 4 \times 10^{-3}$, $V_{td} \sim 8 \times 10^{-3}$.

By using the Cabibbo angle $\lambda \sim 0.22$, we can express the hierarchical structure of V_{CKM} as

$$V_{\mathrm{CKM}} \sim \begin{pmatrix} 1 & \lambda & \lambda^3 \\ \lambda & 1 & \lambda^2 \\ \lambda^3 & \lambda^2 & 1 \end{pmatrix}. \tag{5.13}$$

The CKM matrix satisfies the unitarity condition as

$$V_{\mathrm{CKM}}^\dagger V_{\mathrm{CKM}} = 1. \tag{5.14}$$

Then, we obtain

$$(V_{\rm CKM}^\dagger V_{\rm CKM})_{12} = 0 \qquad \to \qquad V_{ud}^* V_{us} + V_{cd}^* V_{cs} + V_{td}^* V_{ts} = 0, \tag{5.15}$$
$$(V_{\rm CKM}^\dagger V_{\rm CKM})_{13} = 0 \qquad \to \qquad V_{ud}^* V_{ub} + V_{cd}^* V_{cb} + V_{td}^* V_{tb} = 0, \tag{5.16}$$
$$(V_{\rm CKM}^\dagger V_{\rm CKM})_{23} = 0 \qquad \to \qquad V_{us}^* V_{ub} + V_{cs}^* V_{cb} + V_{ts}^* V_{tb} = 0. \tag{5.17}$$

$$\alpha \equiv \arg\left(-\frac{V_{td}V_{tb}^*}{V_{ud}V_{ub}^*}\right), \qquad \beta \equiv \arg\left(-\frac{V_{cd}V_{cb}^*}{V_{td}V_{tb}^*}\right), \qquad \gamma \equiv \arg\left(-\frac{V_{ud}V_{ub}^*}{V_{cd}V_{cb}^*}\right). \tag{5.18}$$

According to PDG [5], the CP violating parameters are given as

$$\sin 2\beta = 0.709 \pm 0.011, \qquad \alpha = (84.1^{+4.5}_{-3.8})^\circ, \qquad \gamma = (65.7 \pm 3.0)^\circ. \tag{5.19}$$

Thus, the CKM matrix has $\mathcal{O}(1)$ CP-violating phase.

5.2.2 Strong CP violation: $\bar{\theta}$

We can define another CP-violating parameter $\bar{\theta}$ as

$$\bar{\theta} = \theta + \arg(\det y_u y_d). \tag{5.20}$$

$\bar{\theta}$ induces the CP violation effect in flavor conserving process, while $\delta_{\rm CKM}$ induces the CP violation effect in flavor changing process. The most important observable to probe $\bar{\theta}$ is the neutron EDM. As shown in eq. (1.4), the neutron EDM is bounded by $|d_n| < 1.8 \times 10^{-26}\, e\,\text{cm}$. Then, by using the result from section 4.4, we obtain the following bound on $\bar{\theta}$:

$$\bar{\theta} \lesssim 10^{-10}. \tag{5.21}$$

5.3 Is up quark massive?

As we have shown in the previous section, there are CP-violating effects in QCD. However, all the observables discussed above are proportional to m_*, which is given by

$$m_* = \left(\frac{1}{m_u} + \frac{1}{m_d} + \frac{1}{m_s}\right)^{-1}. \tag{5.22}$$

Thus, if there exists a massless quark, $m_* = 0$ and all the CP-violating effects in QCD vanish. However, the lightest quark, i.e., the up quark has a nonzero mass $m_u = 2.16 \pm 0.04$ MeV [5].[4] See also ref. [37]. This nonzero quark mass is obtained to explain the observed spectrum of hadrons. Thus, we cannot solve the strong CP problem by the existence of massless quark.

[4]This quark mass is defined in the $\overline{\rm MS}$ scheme at 2 GeV.

5.4 Summary of this section

In the previous section, we have seen that there is the strong CP problem in QCD. In this section, we have discussed the strong CP problem in the Standard Model. Indeed, the situation becomes worse in the Standard Model. It has been confirmed that the CP-violating phase in the CKM matrix is $\mathcal{O}(1)$. Hence, we cannot impose CP symmetry on the Standard Model! Before going to the BSM solutions for the strong CP problem, we reformulate the strong CP problem in a language of topological susceptibility in the next section.

6 Topological susceptibility and the strong CP problem

So far, we have seen that there exists the strong CP problem in QCD, and the situation becomes worse in the Standard Model. In this section, we reformulate the strong CP problem in a language of topological susceptibility [26]. This formulation is also useful to classify possible beyond the Standard Model (BSM) solutions for the strong CP problem. The following papers are useful to understand this section.

- Shifman-Vainshtein-Zakharov [26]
- Gabadadze-Shifman [38] §2
- Pospelov-Ritz [39] §2

6.1 Topological susceptibility

The partition function of QCD with θ term is given by

$$Z = \int \prod_{\phi=A,\psi,\bar{\psi}} \mathcal{D}\phi \exp\left(i\int d^4x \left(\mathcal{L}_{\rm QCD} + \frac{\theta g_s^2}{32\pi^2} G^a_{\mu\nu}\tilde{G}^{a\mu\nu}\right)\right). \tag{6.1}$$

It can be written as

$$Z = \exp\left(-iVTE(\theta)\right), \tag{6.2}$$

where VT is four-dimensional volume and $E(\theta)$ is the vacuum energy density of θ-vacuum.

We can see that θ can be regarded as a source term for $G\tilde{G}$ operator. Then,

$$\langle 0| \frac{g_s^2}{32\pi^2} G^a_{\mu\nu}\tilde{G}^{a\mu\nu} |0\rangle = \frac{-i}{VT}\frac{\partial \log Z}{\partial\theta} = -\frac{\partial E}{\partial\theta}. \tag{6.3}$$

Let us also define the following two-point correlation function:

$$\chi(p^2) \equiv -i\int d^4x e^{ipx} \langle 0|T\ \frac{g_s^2}{32\pi^2} G^a_{\mu\nu}\tilde{G}^{a\mu\nu}(x)\ \frac{g_s^2}{32\pi^2} G^b_{\rho\sigma}\tilde{G}^{b\rho\sigma}(0)\ |0\rangle. \tag{6.4}$$

Zero-momentum limit of $\chi(p^2)$, $\chi(0)$, is called the topological susceptibility. By using the definition of $E(\theta)$, we can also write $\chi(0)$ as

$$\chi(0) = \frac{i}{VT}\frac{\partial^2}{\partial\theta^2}\log Z = \frac{\partial^2 E}{\partial\theta^2}. \tag{6.5}$$

For small θ,

$$\left.\frac{\partial E}{\partial \theta}\right|_{\theta} \simeq \theta \left.\frac{\partial^2 E}{\partial \theta^2}\right|_{\theta=0}, \tag{6.6}$$

which means

$$\langle 0|\frac{g_s^2}{32\pi^2} G^a_{\mu\nu}\tilde{G}^{a\mu\nu}|0\rangle|_\theta \simeq -\theta\chi(0)|_{\theta=0}. \tag{6.7}$$

Thus, nonzero topological susceptibility is a signature of the strong CP problem.

6.2 Shifman-Vainshtein-Zakharov

Let us evaluate $\chi(0)$ as Shifman-Vainshtein-Zakharov did in [26]. For simplicity, we consider two-flavor QCD with u and d quarks. The axial current conservation law is broken by the chiral anomaly and quark mass terms as

$$\partial_\mu J^\mu_A - 2im_u\bar{u}\gamma_5 u - 2im_d\bar{d}\gamma_5 d = -\frac{2g_s^2}{16\pi^2} G^a_{\mu\nu}\tilde{G}^{a\mu\nu}. \tag{6.8}$$

Then, the topological susceptibility can be written as

$$4\chi(0) = \chi_1 + \chi_2. \tag{6.9}$$

where χ_1 and χ_2 are defined as

$$\chi_1 = -\frac{i}{4}\lim_{p\to 0}\int d^4x e^{ipx}\langle 0|(\partial_\mu J^\mu_A)(x)\left(\partial_\mu J^\mu_A - 4i\sum_q m_q\bar{q}\gamma_5 q\right)|0\rangle, \tag{6.10}$$

$$\chi_2 = -i\sum_{q,q'}\lim_{p\to 0}\int d^4x e^{ipx}\langle 0|\left(im_q\bar{q}\gamma^5 q\right)(x)\left(im_{q'}\bar{q}'\gamma^5 q'\right)(0)|0\rangle. \tag{6.11}$$

Note that χ_1 can be rewritten as

$$\begin{aligned}
\chi_1 &= -\frac{i}{4}\lim_{p\to 0}\int d^4x e^{ipx}\langle 0|(\partial_\mu J^\mu_A)(x)\left(\frac{-2g_s^2}{16\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu} - 2i\sum_q m_q\bar{q}\gamma_5 q\right)|0\rangle, \\
&= \frac{i}{4}\lim_{p\to 0}\int d^4x e^{ipx}\delta(x^0)\langle 0|\left[J^0_A(x), \left(-2i\sum_q m_q\bar{q}\gamma_5 q\right)\right]|0\rangle, \qquad (6.12)\\
&= -\sum_q\langle 0|m_q\bar{q}q|0\rangle. \qquad (6.13)
\end{aligned}$$

Then, we can see that $\chi(0)$ can be written by quark condensate and pseudoscalar susceptibility as [40–42]

$$4\chi(0) = -\sum_q\langle 0|m_q\bar{q}q|0\rangle - i\sum_{q,q'}\lim_{p\to 0}\int d^4x e^{ipx}\langle 0|\left(im_q\bar{q}\gamma^5 q\right)(x)\left(im_{q'}\bar{q}'\gamma^5 q'\right)(0)|0\rangle. \tag{6.14}$$

The first term of RHS is $\mathcal{O}(m_q)$. The second term of RHS is naively $\mathcal{O}(m_q^2)$, but it can be enhanced by one particle state whose mass is $\mathcal{O}(\sqrt{m_q})$. In this case, the possible candidate of such a particle is π^0 meson.

The quark condensate can be evaluated as [43–45]

$$\sum_q \langle 0|m_q \bar{q} q|0\rangle \simeq -f_\pi^2 m_\pi^2 \tag{6.15}$$

This is so-called Gell-Mann–Oakes–Renner relation. For the second term,[5]

$$\langle 0|\bar{u}\gamma^5 u|\pi^0\rangle = -\langle 0|\bar{d}\gamma^5 d|\pi^0\rangle = \frac{f_\pi m_\pi^2}{m_u + m_d}, \tag{6.18}$$

Then, we obtain

$$\begin{aligned}\chi(0) &\simeq \frac{1}{4} f_\pi^2 m_\pi^2 - \frac{i}{4}\left(\frac{f_\pi m_\pi^2}{m_u + m_d}(m_u - m_d)\right)^2 \frac{i}{-m_\pi^2} \\ &= f_\pi^2 m_\pi^2 \frac{m_u m_d}{(m_u + m_d)^2}.\end{aligned} \tag{6.19}$$

Naively, the first term and the second term in eq. (6.14) are $\mathcal{O}(m_q)$ and $\mathcal{O}(m_q^2)$, respectively. Thus, the second term can cancel the first term only if one particle state whose mass is $\mathcal{O}(\sqrt{m_q})$ contributes to the second term. Therefore, it is clear that $\chi(0)$ is nonzero if there is no pseudoscalar meson with $m^2 \sim \mathcal{O}(m_q)$ other than π^0. Also, we can see that π^0 pole contribution exactly cancels the first term if either $m_u = 0$ or $m_d = 0$.

6.3 How can we solve the strong CP problem?

Eq. (6.7) implies solutions of the strong CP problem can be classified to the following two categories:

- $\chi(0) = 0$ (or more precisely $E(\theta)$ is θ-independent) because of BSM particle(s)
- $\chi(0) \neq 0$ but UV physics explains the reason of $\bar{\theta} \simeq 0$.

6.4 Summary of this section

In this section, we formulate the strong CP problem and $U(1)_A$ problem in terms of the topological susceptibility. We have seen that nonzero topological susceptibility is a signature of the strong CP problem, and it is closely connected to the mass of η' meson. (for details, see Appendix A) This implies that the strong CP problem in the Standard Model is an inevitable consequence of the observed mass spectrum of pseudoscalar mesons,

[5] From the definition of f_π,

$$\langle 0|\bar{u}\gamma^5\gamma^\mu d|\pi^+\rangle = -\sqrt{2} i f_\pi p^\mu. \tag{6.16}$$

Then, by using the equation of motion, we can obtain

$$\langle 0|\bar{u}\gamma^5 d|\pi^+\rangle = \sqrt{2} f_\pi m_\pi^2/(m_u + m_d). \tag{6.17}$$

together with the neutron EDM bound. Also, this formulation is useful to classify possible BSM solutions for the strong CP problem. In the next two sections, we will discuss two well-known solutions to the strong CP problem, i.e., the QCD axion and the Nelson-Barr mechanism.

7 QCD axion

In this section, we will discuss the QCD axion, which is one of the most beautiful solutions to the strong CP problem. Peccei and Quinn [3, 4] proposed a solution to the strong CP problem by introducing a new global $U(1)$ symmetry, which is called Peccei-Quinn (PQ) symmetry. Then, Weinberg [46] and Wilczek [47] pointed out that this solution predicts the existence of a new light particle, which is called the QCD axion. The QCD axion is a pseudo-Nambu-Goldstone boson associated with the spontaneous breaking of PQ symmetry, and it has a coupling to gluons. The following papers are useful to understand this section.

- Kim-Carosi [48]
- Kawasaki-Nakayama [49]
- Hook [50]
- Caputo-Raffelt [51]

7.1 Axion EFT

Let us consider a scalar field a with the following Lagrangian:

$$\mathcal{L}_a = \frac{1}{2}(\partial_\mu a)(\partial^\mu a) + \frac{a}{f_a}\frac{g_s^2}{32\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu}. \tag{7.1}$$

We define the shift symmetry of a as

$$a \to a + \delta, \tag{7.2}$$

where δ is a constant. The kinetic term is invariant under this shift symmetry. The coupling between a and gluons is not invariant, but it is invariant up to a total derivative term. Thus, under this shift symmetry, the Lagrangian is invariant at the classical level. However, it is not invariant at the quantum level. At the quantum level, the theory is invariant under the finite shift of a as $a \to a + 2\pi f_a$.

The axion Lagrangian with the θ term is given by

$$\mathcal{L} = \frac{1}{2}(\partial_\mu a)(\partial^\mu a) + \left(\bar{\theta} + \frac{a}{f_a}\right)\frac{g_s^2}{32\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu}. \tag{7.3}$$

After integrating out the quark and gluon field in the path integral, we obtain the following effective Lagrangian:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu a)(\partial^\mu a) - V\left(\bar{\theta} + \frac{a}{f_a}\right). \tag{7.4}$$

7.1.1 Vafa-Witten theorem

Let us write the potential $V(\theta)$ by using the Euclidean path integral as

$$V(\theta) = -\frac{1}{V_4}\log\int \mathcal{D}A_\mu \mathcal{D}q\mathcal{D}\bar{q}\exp\left(-\int d^4x(\mathcal{L}_{\rm QCD} + \frac{i\theta g_s^2}{32\pi^2}G_{ij}^a\tilde{G}_{ij}^a)\right), \tag{7.5}$$

where V_4 is the four-dimensional volume. Note that i and j run over $1,2,3,4$ in Euclidean space. Integrating out the quark field, we obtain

$$V(\theta) = -\frac{1}{V_4}\log\int \mathcal{D}A_\mu \det(\not{D}+M)\exp\left(-\int d^4x\frac{1}{4}G_{ij}^aG_{ij}^a\right)\times\exp\left(-\int d^4x\frac{i\theta g_s^2}{32\pi^2}G_{ij}^a\tilde{G}_{ij}^a\right). \tag{7.6}$$

In vector-like gauge theory such as QCD, the fermion determinant $\det(\not{D}+M)$ is positive definite for positive real quark masses [52]. Thus, the θ-independent part of integrand is positive definite. Thus, we conclude that

$$V(\theta) \geq V(0). \tag{7.7}$$

This is the Vafa-Witten theorem [53]. At the potential minimum of V, we obtain

$$\bar{\theta} + \frac{\langle a\rangle}{f_a} = 0. \tag{7.8}$$

Thus, the strong CP problem is dynamically solved by the axion field a![6] The mass of the axion is evaluated as

$$m_a^2 = \frac{1}{f_a^2}\frac{\partial^2 V}{\partial\theta^2}\bigg|_{\theta=0} \tag{7.9}$$

Since $\partial^2 V/\partial\theta^2|_{\theta=0}$ is nothing but the topological susceptibility χ of QCD, we obtain

$$m_a^2 = \frac{\chi_{\rm QCD}}{f_a^2}, \tag{7.10}$$

where

$$\chi_{\rm QCD} = f_\pi^2 m_\pi^2\frac{m_u m_d}{(m_u+m_d)^2}. \tag{7.11}$$

7.1.2 Axion in the chiral Lagrangian

To discuss physics of the axion whose Lagrangian is given in eq. (7.3), we can implement the axion field a in the chiral Lagrangian given in eq. (4.49), which can be obtained by replacing $\bar{\theta} \to \bar{\theta} + a/f_a$. Actually, we can absorb $\bar{\theta}$ into the definition of a. Then, we can simply replace $\bar{\theta} \to a/f_a$ to obtain the axion potential.

Here, we consider the two-flavor case for simplicity. Then, we can take the following quark mass matrix,

$$M = {\rm diag}(m_u\exp\left(\frac{im_*a}{m_uf_a}\right), m_d\exp\left(\frac{im_*a}{m_df_a}\right)), \tag{7.12}$$

[6] CKM phase can induce a non-zero neutron EDM but this is of the order of $10^{-31} - 10^{-33}$ e cm [54].

where $m_* = (1/m_u + 1/m_d)^{-1}$. By focusing on the neutral pion π^0, we can take the following form of Σ:

$$\Sigma = \mathrm{diag}(\exp\left(\frac{i\pi^0}{f_\pi}\right), \exp\left(-\frac{i\pi^0}{f_\pi}\right)). \tag{7.13}$$

Then, the potential of π^0 and a is given by

$$\begin{aligned}
V &= -\frac{1}{2} f_\pi^2 B_0 \mathrm{tr}[M\Sigma + M^\dagger \Sigma^\dagger] \\
&= -f_\pi^2 B_0 \left(m_u \cos\left(\frac{m_* a}{m_u f_a} + \frac{\pi^0}{f_\pi}\right) + m_d \cos\left(\frac{m_* a}{m_d f_a} - \frac{\pi^0}{f_\pi}\right)\right) \\
&= -f_\pi^2 B_0 (m_u + m_d) + \frac{1}{2} f_\pi^2 B_0 \left(\frac{1}{m_u} + \frac{1}{m_d}\right) m_*^2 \frac{a^2}{f_a^2} + \frac{1}{2}(m_u + m_d) B_0 (\pi^0)^2 + \cdots \\
&= -f_\pi^2 B_0 (m_u + m_d) + \frac{1}{2} f_\pi^2 B_0 m_* \frac{a^2}{f_a^2} + \frac{1}{2}(m_u + m_d) B_0 (\pi^0)^2 + \cdots .
\end{aligned} \tag{7.14}$$

This reproduces the relation $m_\pi^2 = B_0(m_u + m_d)$ again. In addition, we can also obtain the mass of axion as

$$m_a^2 = \left.\frac{\partial^2 V}{\partial a^2}\right|_{a=0} = \frac{f_\pi^2 B_0 m_*}{f_a^2} = \frac{f_\pi^2 m_\pi^2}{f_a^2} \frac{m_u m_d}{(m_u + m_d)^2}. \tag{7.15}$$

We can easily see that this is consistent with eq. (7.10). By using $m_u = 2.16$ MeV and $m_d = 4.70$ MeV [5], we can evaluate the axion mass as

$$m_a \simeq 5.8\ \mu\mathrm{eV} \times \left(\frac{10^{12}\ \mathrm{GeV}}{f_a}\right). \tag{7.16}$$

7.2 UV completion of QCD axion

The QCD axion has shift symmetry. Then, interpreting the QCD axion as a NG boson from the spontaneous breaking of a global $U(1)$ symmetry is a natural idea. This symmetry is called Peccei-Quinn (PQ) symmetry. To obtain the coupling between the axion and gluons, this symmetry should be anomalous under $SU(3)_C$ gauge symmetry. Thus, we need to assign PQ charge to colored particles in a chiral way. There are two possible simple options:

- The SM quarks do not have PQ charge: KSVZ axion model
- The SM quarks have PQ charge: PQWW, DFSZ axion model

7.2.1 KSVZ model

One of the simplest UV completions of QCD axion is proposed by Kim [55] and Shifman, Vainshtein, and Zakharov [26]. This model is called the KSVZ axion model. Let us introduce a new complex scalar field Φ and a vector-like quark $Q_{L,R}$, which are charged under $U(1)_{\mathrm{PQ}}$ symmetry. The matter content of KSVZ model is shown in table 5. The Lagrangian of KSVZ model is given by

$$\mathcal{L} = -\lambda \left(|\Phi|^2 - \frac{v_\Phi^2}{2}\right)^2 - \left(y\Phi \bar{Q}_R Q_L + \mathrm{h.c.}\right). \tag{7.17}$$

	$SU(3)_C$	$U(1)_{\rm PQ}$
Q_L	3	0
Q_R	3	1
Φ	1	1

Table 5. Matter content of KSVZ model.

At the vacuum, Φ obtains a VEV as $\langle\Phi\rangle = v_\Phi/\sqrt{2}$. Then, the PQ symmetry is spontaneously broken, and the axion is identified as the phase of Φ:

$$\Phi = \frac{v_\Phi}{\sqrt{2}} e^{ia/v_\Phi}. \tag{7.18}$$

To eliminate the axion field dependence in the mass term of fermion Q, we can perform the following chiral rotation of Q_R:

$$Q_L(x) \to Q_L(x), \qquad Q_R(x) \to \exp\left(\frac{ia(x)}{v_\Phi}\right) Q_R(x). \tag{7.19}$$

Then, we obtain the following Lagrangian for a and Q:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu a)(\partial^\mu a) + \frac{a}{v_\Phi}\frac{g_s^2}{32\pi^2} G^a_{\mu\nu}\tilde{G}^{a\mu\nu} - \frac{1}{v_\Phi}(\partial_\mu a)\bar{Q}\gamma^\mu P_R Q - \frac{y v_\Phi}{\sqrt{2}}\bar{Q}Q. \tag{7.20}$$

By integrating out the heavy fermion Q, we can drop the third term and the last term. Then, we obtain the effective Lagrangian of the axion as given in eq. (7.1). Comparing eq. (7.20) with eq. (7.1), we can identify the axion decay constant f_a as

$$f_a = v_\Phi. \tag{7.21}$$

7.2.2 PQWW, DFSZ model

	$SU(3)_C$	$U(1)_{\rm PQ}$
q_L	3	0
u_R	3	1
d_R	3	0
ℓ_L	1	0
e_R	1	0
H_1	1	0
H_2	1	1

	$SU(3)_C$	$U(1)_{\rm PQ}$
q_L	3	0
u_R	3	1
d_R	3	0
ℓ_L	1	0
e_R	1	-1
H_1	1	0
H_2	1	1

Table 6. Matter content of PQWW-I model (left) and PQWW-II model (right).

Let us assume the SM quarks have PQ charges. Then, the Higgs field also has nonzero PQ charge. The minimal Higgs sector with PQ symmetry is two Higgs doublet model. This axion model is called PQWW model [46, 47]. The Higgs potential and Yukawa interaction of PQWW model are given by

$$V = m_{11}^2|H_1|^2 + m_{22}^2|H_2|^2 + \frac{\lambda_1}{2}|H_1|^4 + \frac{\lambda_2}{2}|H_2|^4 + \lambda_3|H_1|^2|H_2|^2 + \lambda_4|H_1^\dagger H_2|^2. \tag{7.22}$$

The quark Yukawa interaction is given by

$$\mathcal{L}_{\text{yuk}} = -y_{u,ij}\bar{u}_{R,i}\tilde{H}_2^\dagger q_{L,j} - y_{d,ij}\bar{d}_{R,i}H_1^\dagger q_{L,j} + \text{h.c.}. \tag{7.23}$$

For the lepton Yukawa interaction, there are two choices:

$$\mathcal{L}_{(\text{I})} = -y_{e,ij}\bar{e}_{R,i}H_1^\dagger \ell_{L,j} + \text{h.c.}, \tag{7.24}$$

$$\mathcal{L}_{(\text{II})} = -y_{e,ij}\bar{e}_{R,i}H_2^\dagger \ell_{L,j} + \text{h.c.} \tag{7.25}$$

Here we call the first choice as PQWW-I and the second choice as PQWW-II. See also table 6.

Let us decompose H_1 and H_2 as

$$H_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v_1 \end{pmatrix} e^{ia_1/v_1}, \qquad H_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v_2 \end{pmatrix} e^{ia_2/v_2}. \tag{7.26}$$

$v_1 = v\cos\beta$ and $v_2 = v\sin\beta$ with $v = 246$ GeV.

$$\begin{aligned}\mathcal{L} =& \frac{1}{2}(\partial_\mu a_1 - \frac{1}{2}g_Z Z_\mu v_1)^2 + \frac{1}{2}(\partial_\mu a_2 - \frac{1}{2}g_Z Z_\mu v_2)^2 \\ =& \frac{1}{2}(\frac{1}{2}g_Z v Z_\mu - \cos\beta\partial_\mu a_1 - \sin\beta\partial_\mu a_2)^2 + \frac{1}{2}(\partial_\mu a_1)^2 + \frac{1}{2}(\partial_\mu a_2)^2 - \frac{1}{2}(\cos\beta\partial_\mu a_1 + \sin\beta\partial_\mu a_2)^2 \\ =& \frac{1}{2}(\frac{1}{2}g_Z v Z_\mu - \cos\beta\partial_\mu a_1 - \sin\beta\partial_\mu a_2)^2 + \frac{1}{2}(\partial_\mu(a_1\sin\beta - a_2\cos\beta))^2, \end{aligned} \tag{7.27}$$

where $g_Z \equiv \sqrt{g'^2 + g^2}$, $\tan\theta_W = g'/g$, and $Z_\mu \equiv \cos\theta_W W_\mu^3 - \sin\theta_W B_\mu$. Then, we define a_H and G as

$$a_H = -a_1\sin\beta + a_2\cos\beta, \qquad G = a_1\cos\beta + a_2\sin\beta. \tag{7.28}$$

G is eaten as a longitudinal mode of Z boson, and a_H is identified as the axion.

$$a_1 = -a_H\sin\beta + G\cos\beta, \qquad a_2 = a_H\cos\beta + G\sin\beta. \tag{7.29}$$

H_1 and H_2 can be rewritten as

$$H_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v_1 \end{pmatrix} \exp\left(i\frac{-a_H\tan\beta}{v} + \frac{iG}{v}\right), \qquad H_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v_2 \end{pmatrix} \exp\left(i\frac{a_H\cot\beta}{v} + \frac{iG}{v}\right). \tag{7.30}$$

Let us absorb a_H by the following chiral rotation of quark fields:

$$u_R \to \exp\left(i\frac{a_H\cot\beta}{v}\right)u_R, \qquad d_R \to \exp\left(i\frac{a_H\tan\beta}{v}\right)d_R. \tag{7.31}$$

Then, we obtain the following Lagrangian for a_H:

$$\mathcal{L} = \frac{3a_H}{f_v}\frac{g_s^2}{32\pi^2}G_{\mu\nu}^a\tilde{G}^{a\mu\nu} - \frac{\cos^2\beta}{f_v}(\partial_\mu a_H)\bar{u}\gamma^\mu P_R u - \frac{\sin^2\beta}{f_v}(\partial_\mu a_H)\bar{d}\gamma^\mu P_R d. \tag{7.32}$$

	$SU(3)_C$	$U(1)_{\rm PQ}$
q_L	3	0
u_R	3	2
d_R	3	0
ℓ_L	1	0
e_R	1	0
H_1	1	0
H_2	1	2
S	1	-1

	$SU(3)_C$	$U(1)_{\rm PQ}$
q_L	3	0
u_R	3	2
d_R	3	0
ℓ_L	1	0
e_R	1	-2
H_1	1	0
H_2	1	2
S	1	-1

Table 7. Matter content of DFSZ-I model (left) and DFSZ-II model (right).

where f_v is given by

$$f_v = v \sin\beta \cos\beta. \tag{7.33}$$

However, PQWW axion model is already excluded because of too low $f_v \sim \mathcal{O}(100)$ GeV [56, 57].

DFSZ model [58, 59] is a minimal extension of PQWW model, in which we introduce a complex scalar field S with PQ charge. We can write the following renormalizable scalar potential of DFSZ model:[7]

$$V_S = \lambda_S \left(|S|^2 - \frac{v_S^2}{2}\right)^2 + \lambda_{SH}(S^2 H_1^\dagger H_2 + \text{h.c.}). \tag{7.35}$$

Let us write S as

$$S = \frac{v_S}{\sqrt{2}} e^{i a_s / v_S}. \tag{7.36}$$

Then, the potential terms in eq. (7.35) can be rewritten as

$$\lambda_{SH}(S^2 H_1^\dagger H_2 + \text{h.c.}) \to \lambda_{SH} \frac{v_S^2 v^2}{2} \sin\beta\cos\beta\cos\left(2\frac{a_s}{v_S} + \frac{a_H}{f_v}\right), \tag{7.37}$$

respectively. This term gives mass to A, which is a linear combination of a_s and a_H. The massless mode a, orthogonal to the massive mode, is identified as

$$a \simeq -a_s + \frac{2f_v}{v_S} a_H. \tag{7.38}$$

Inversely, we obtain

$$a_s \simeq -a + \frac{2f_v}{v_S} A, \qquad a_H \simeq A + a\frac{2f_v}{v_S}. \tag{7.39}$$

[7]Note that we could also use the following potential term instead of eq. (7.35):

$$V_S = \lambda_S \left(|S|^2 - \frac{v_S^2}{2}\right)^2 + \kappa(S H_1^\dagger H_2 + \text{h.c.}). \tag{7.34}$$

In this case, the PQ charge assignment becomes different from the one in table 7, and we obtain $N_{\rm DW} = 3$.

Similar to the PQWW model, the Yukawa interaction of quarks obtains the axion dependence and this can be eliminated by the following chiral rotation of quark fields:

$$u_R \to \exp\left(i\frac{2a\cos^2\beta}{v_S}\right)u_R, \qquad d_R \to \exp\left(i\frac{2a\sin^2\beta}{v_S}\right)d_R. \tag{7.40}$$

Finally, we obtain the following Lagrangian for the axion:

$$\mathcal{L} = \frac{6a}{v_S}\frac{g_s^2}{32\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu} - \frac{2\cos^2\beta}{v_S}(\partial_\mu a)\bar{u}\gamma^\mu P_R u - \frac{2\sin^2\beta}{v_S}(\partial_\mu a)\bar{d}\gamma^\mu P_R d. \tag{7.41}$$

By comparing this with eq. (7.1), we can identify the axion decay constant f_a as

$$f_a = \frac{v_S}{6}. \tag{7.42}$$

7.3 Domain wall number

So far we have seen several explicit models of QCD axion. In low energy effective description, the axion has a coupling to gluons as

$$\mathcal{L} = \frac{a}{f_a}\frac{g_s^2}{32\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu}. \tag{7.43}$$

The Vafa-Witten theorem tells us that the potential minimum is at $a = 0$. However, the periodicity of $\bar{\theta}$ tells us that $a = 2\pi n f_a$ ($n \in \mathbb{Z}$) are also potential minima. Does it mean the axion potential has multiple minima? The answer depends on the model.

In the KSVZ axion model, the axion is identified as the phase of Φ:

$$\Phi = \frac{f_a}{\sqrt{2}}e^{ia/f_a}. \tag{7.44}$$

Thus, the periodicity of a is given by

$$a \sim a + 2\pi f_a. \tag{7.45}$$

This means $a = 0$ and $a = 2\pi f_a$ are the same point in the field space. Thus, there is only one potential minimum.

In the DFSZ axion model, the axion is identified as the phase of S:

$$S = \frac{6f_a}{\sqrt{2}}e^{ia/6f_a}, \tag{7.46}$$

Thus, the periodicity of a is given by

$$a \sim a + 12\pi f_a. \tag{7.47}$$

and there exists six degenerate potential minima:

$$a = 0,\ 2\pi f_a,\ 4\pi f_a,\ 6\pi f_a,\ 8\pi f_a,\ 10\pi f_a. \tag{7.48}$$

The number of physical global minima of the axion potential is called domain wall number $N_{\rm DW}$, i.e.,

$$N_{\rm DW}|_{\rm KSVZ} = 1, \qquad N_{\rm DW}|_{\rm DFSZ} = 6. \tag{7.49}$$

Note that $U(1)_{\rm PQ}$ is explicitly broken by QCD interaction, but $Z_{N_{\rm DW}}$ subgroup of $U(1)_{\rm PQ}$ is unbroken because of the degeneracy of the potential minima. Thus, DFSZ has Z_6 symmetry. However, both of them are spontaneously broken by the VEV of S.

Suppose that the PQ symmetry is not broken at the end of inflation, i.e., $\langle \Phi \rangle = 0$ in KSVZ or $\langle S \rangle = 0$ in DFSZ. First, the PQ symmetry is spontaneously broken at the PQ phase transition. Then, the axion obtains a VEV. This phase transition should be before QCD phase transition. Then, the QCD nonperturbative effect is negligible and the axion potential is flat. Thus, we can understand there is no explicit breaking of $U(1)_{\rm PQ}$, but it is spontaneously broken. From this symmetry breaking, we can expect the formation of cosmic strings. After the QCD phase transition, the axion obtains a potential and $U(1)_{\rm PQ}$ is explicitly broken to $Z_{N_{\rm DW}}$. Then, $N_{\rm DW}$ degenerate vacua appear. Thus, we have domain wall if $N_{\rm DW} > 1$. $N_{\rm DW}$ is attached to the cosmic string. Such a string-domain wall network could overclose the universe.

7.4 Generic axion EFT

The EFT given in eq. (7.1) is a minimal setup of the axion to solve the strong CP problem. However, as we have seen so far, in general, the axion can couple to other SM particles; quarks, leptons, and electroweak gauge bosons. Here we discuss a generic axion EFT below the electroweak scale. The generic axion EFT is given by

$$\mathcal{L} = \frac{1}{2}(\partial_\mu a)(\partial^\mu a) + \frac{a}{f_a}\frac{g_s^2}{32\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu} + \frac{E}{2N}\frac{a}{f_a}\frac{e^2}{16\pi^2}F_{\mu\nu}\tilde{F}^{\mu\nu} - \frac{1}{f_a}(\partial_\mu a) j^\mu_{\rm PQ}, \tag{7.50}$$

where $F_{\mu\nu}$ is the field strength of the photon and $j_{\rm PQ}$ is the PQ current of SM fermions. The anomaly coefficients N and E are given by

$$N = \sum_f (Q^{f_R}_{\rm PQ} - Q^{f_L}_{\rm PQ}) C(r), \qquad E = \sum_f (Q^{f_R}_{\rm PQ} - Q^{f_L}_{\rm PQ})(Q^f_{\rm em})^2 d(r), \tag{7.51}$$

$C(r)$ is the Dynkin index of the representation r defined by

$$\mathrm{tr}[T^a_r T^b_r] = C(r)\delta^{ab}. \tag{7.52}$$

For example, $C(\text{fund}) = 1/2$ and $C(\text{adj}) = N$ for $SU(N)$ gauge group. $d(r)$ is the dimension of the representation r. $Q^f_{\rm em}$ and $Q^{f_{L,R}}_{\rm PQ}$ are the electric and PQ charges of fermion f, respectively. The sum is taken over all fermions in the theory. With the normalization in which the coefficient of $aG\tilde{G}$ is $1/f_a$, the corresponding fermion current $j^\mu_{\rm PQ}$ is

$$j^\mu_{\rm PQ} = \frac{1}{2N}\sum_f \bar{f}\gamma^\mu (Q^{f_L}_{\rm PQ} P_L + Q^{f_R}_{\rm PQ} P_R) f. \tag{7.53}$$

For the minimal KSVZ axion model, $E/N = 0$ and $j^\mu_{\rm PQ} = 0$. For the DFSZ-I axion model, The factors in the following expressions are the chiral PQ charge, the group-theory factor, and the multiplicities:

$$N = \underbrace{2}_{Q_{\rm PQ}} \times \underbrace{\frac{1}{2}}_{C(\mathbf{3})} \times \underbrace{3}_{N_{\rm gen}} = 3, \tag{7.54}$$

$$E = \underbrace{2}_{Q_{\rm PQ}} \times \underbrace{\left(\frac{2}{3}\right)^2}_{Q^2_{\rm em}} \times \underbrace{3}_{N_c} \times \underbrace{3}_{N_{\rm gen}} = 8, \tag{7.55}$$

therefore

$$\frac{E}{N} = \frac{8}{3}. \tag{7.56}$$

The PQ current is given by

$$j^\mu_{\rm PQ} = \frac{1}{3}\cos^2\beta \bar{u}\gamma^\mu P_R u + \frac{1}{3}\sin^2\beta \bar{d}\gamma^\mu P_R d + \frac{1}{3}\sin^2\beta \bar{e}\gamma^\mu P_R e. \tag{7.57}$$

Note that this PQ charge is different from the original PQ charge assignment because the axion a is chosen to be orthogonal to the longitudinal mode of Z boson. This PQ charge should be understood as $Q_{\rm PQ} - 4\sin^2\beta(Y - B/2 + L/2)$. For the DFSZ-II axion model,

$$N = \underbrace{2}_{Q_{\rm PQ}} \times \underbrace{\frac{1}{2}}_{C(\mathbf{3})} \times \underbrace{3}_{N_{\rm gen}} = 3, \tag{7.58}$$

$$E = \left[\underbrace{2}_{Q_{\rm PQ}} \times \underbrace{\left(\frac{2}{3}\right)^2}_{Q^2_{\rm em}} \times \underbrace{3}_{N_c} + \underbrace{(-2)}_{Q_{\rm PQ}} \times \underbrace{(-1)^2}_{Q^2_{\rm em}}\right] \times \underbrace{3}_{N_{\rm gen}} = 2, \tag{7.59}$$

therefore,

$$\frac{E}{N} = \frac{2}{3}. \tag{7.60}$$

The PQ current is given by

$$j^\mu_{\rm PQ} = \frac{1}{3}\cos^2\beta \bar{u}\gamma^\mu P_R u + \frac{1}{3}\sin^2\beta \bar{d}\gamma^\mu P_R d - \frac{1}{3}\cos^2\beta \bar{e}\gamma^\mu P_R e. \tag{7.61}$$

Again, this PQ charge is different from the original PQ charge assignment because the axion a is chosen to be orthogonal to the longitudinal mode of Z boson. This PQ charge should be understood as $Q_{\rm PQ} - 4\sin^2\beta(Y - B/2 + L/2)$.

After integrating out mesons, the axion-photon coupling is given by

$$\mathcal{L} = \frac{1}{4} g_{a\gamma\gamma} a F_{\mu\nu}\tilde{F}^{\mu\nu}, \tag{7.62}$$

where $g_{a\gamma\gamma}$ is given by

$$g_{a\gamma\gamma} = \frac{\alpha_{\rm em}}{2\pi f_a}\left(\frac{E}{N} - \Delta\right), \tag{7.63}$$

where Δ is $(2/3)(4m_d + m_u)/(m_d + m_u) \simeq 2.04$ at the LO chiral perturbation theory, and 1.92 at NLO chiral perturbation theory [60].

7.5 Where is the axion? — How large is f_a?

Here we briefly summarize the phenomenology of QCD axion and the constraints on f_a.

7.5.1 Lower bound on f_a

The axion decay constant f_a should be sufficiently large, as $f_a \gtrsim 10^8$–10^9 GeV. Here we briefly look at several constraints on f_a. Here we are sloppy about the model dependence of the axion couplings to SM particles. Each bound on f_a should be understood up to model-dependent factors of order unity.

$K^+ \to \pi^+ a$

One of the most important laboratory constraints on the axion comes from the rare kaon decay, in particular, $K^+ \to \pi^+ a$. The rare kaon decay is searched by NA62 [61] as $\mathrm{Br}(K^+ \to \pi^+ X) < 3 \times 10^{-11}$ (90% C.L.) for $m_X \simeq 0$ and $\tau = +\infty$. This constraint can be translated into the lower bound on f_a as $f_a \gtrsim 1$ TeV. (See, e.g., ref. [62].)

Stellar cooling

Axions can be produced in stellar interiors and escape from the star, providing an additional cooling channel. For example, the helium-burning lifetime of horizontal-branch stars constrains the axion-photon coupling as $g_{a\gamma\gamma} \lesssim 0.65 \times 10^{-10}$ GeV^{-1} (95% C.L.) [51]. This bound can be translated into $f_a \gtrsim 3.4 \times 10^7$ GeV for the minimal KSVZ axion [51].

SN1987A

Axion emission provides an additional cooling channel for supernovae and can shorten the duration of the neutrino burst. The measurement of neutrino signals from SN1987A constrains $f_a \gtrsim 4 \times 10^8$ GeV for the minimal KSVZ axion [51]. For more details on the discussion of supernovae (including theoretical uncertainties), see §8 of ref. [51].

7.5.2 Axion dark matter: misalignment mechanism

$f_a \sim 10^{12}$ GeV is interesting since the observed dark matter abundance can be explained by a simple scenario of the axion dark matter.

The equation of motion of the axion in the FRW universe is given by

$$\ddot{a} + 3H\dot{a} + V'(a) \simeq 0. \tag{7.64}$$

The second term in the LHS is the Hubble friction term. Then, the axion starts to roll when

$$3H(T_{\rm osc}) \simeq m_a(T_{\rm osc}), \tag{7.65}$$

where $m_a(T) = \sqrt{\partial^2 V_T / \partial a^2}$ is temperature dependent axion mass. After the onset of oscillation, the coherent oscillation of the axion field behaves as cold dark matter.

Ref. [63] gives the temperature dependent axion mass as

$$m_a^2(T) = m_{a,0}^2 \times \begin{cases} (T/T_c)^{-\gamma} & T > T_c \\ 1 & T < T_c \end{cases}, \tag{7.66}$$

where $T_c = 2.12\times\Lambda_{b,0}$, $\Lambda_{b,0} = 75.6$ MeV, $\gamma = 8.16$. By using $H(T) = \sqrt{\pi^2 g_*(T)/90}T^2/M_{\rm Pl}$, we can estimate $T_{\rm osc}$ as

$$T_{\rm osc} \sim 1.0 \text{ GeV} \times \left(\frac{f_a}{10^{12}\text{ GeV}}\right)^{-0.16} \times \left(\frac{g_*(T_{\rm osc})}{70}\right)^{-0.08}. \tag{7.67}$$

The axion yield can be estimated as

$$Y_a \equiv \frac{n_a}{s} \sim \left.\frac{\rho_a}{m_a s}\right|_{T=T_{\rm osc}} \sim \left.\frac{m_a}{s}\right|_{T=T_{\rm osc}} \times f_a^2\theta_i^2, \tag{7.68}$$

where s is the entropy density of the universe and θ_i is the initial misalignment angle. Then, we can estimate the axion abundance as

$$\Omega_a h^2 = \frac{m_{a,0}n_a}{\rho_c/h^2} = \frac{m_{a,0}Y_a s_0}{\rho_c/h^2} \tag{7.69}$$

Here $s_0 = 2891.2\text{ cm}^{-3}$ is the entropy density of the present universe [5] and $\rho_c = 3H_0^2M_{\rm Pl}^2 = h^2 \times 1.05375 \times 10^{-5}\text{ GeV/cm}^3$ is the critical density of the present universe. Finally, we obtain

$$\Omega_a h^2 \sim 0.2 \times \theta_i^2 \left(\frac{f_a}{10^{12}\text{ GeV}}\right)^{1.16} \left(\frac{g_*(T_{\rm osc})}{70}\right)^{0.58} \left(\frac{g_{*s}(T_{\rm osc})}{70}\right)^{-1}. \tag{7.70}$$

Thus, assuming $\theta_i \sim \mathcal{O}(1)$, we can explain the observed dark matter abundance $\Omega_{\rm DM}h^2 \simeq 0.12$ for $f_a \sim 10^{12}$ GeV. For more details, see, e.g., refs. [49, 64].

7.6 The axion quality problem

The existence of axion requires the existence of a gauge invariant operator $\mathcal{O}_d$ charged under the PQ symmetry, which obtains nonzero VEV. Then, there is no reason to forbid the existence of higher dimensional operator $\mathcal{O}_d$ in the Lagrangian as

$$\mathcal{L} \sim \frac{c}{\Lambda^{d-4}}\mathcal{O}_d. \tag{7.71}$$

Then, the axion potential is modified as

$$V \sim -\chi_{\rm QCD}\cos\frac{a}{f_a} - \frac{\langle\mathcal{O}_d\rangle}{\Lambda^{d-4}}\cos\left(n\frac{a}{f_a} + \arg c\right), \tag{7.72}$$

where n is the PQ charge of $\mathcal{O}_d$. This induces nonzero $\bar\theta$ as

$$\bar\theta \sim \frac{1}{\chi_{\rm QCD}(0)}\frac{\langle\mathcal{O}_d\rangle}{\Lambda^{d-4}}. \tag{7.73}$$

For $f_a \sim 10^{12}$ GeV and $\Lambda \sim M_{\rm pl} \sim 10^{18}$ GeV, $d \gtrsim 13$–14 is required to satisfy $\bar\theta \lesssim 10^{-10}$. This is so-called axion quality problem. For a review see [65]. The possible directions would be

- large d : composite axion
- non-local $\mathcal{O}$: wilson line (or 5th component of gauge field) in 5d model

7.7 Summary of this section

In this section, we have discussed the QCD axion as a solution to the strong CP problem. The QCD axion is a pseudo Nambu-Goldstone boson of the spontaneously broken PQ symmetry, which is anomalous under QCD. The QCD axion absorbs the $\bar{\theta}$ parameter and its potential minimum, and this is guaranteed by the Vafa-Witten theorem. Depending on the details of the model, the axion can couple to other SM particles, and this gives rich phenomenology. The QCD axion can also be a candidate of dark matter.

8 Nelson-Barr mechanism

Here we discuss Nelson-Barr mechanism, which is another solution to the strong CP problem based on spontaneous CP violation [66–68]. For a review, see, e.g., ref. [69]. By imposing CP symmetry on the Lagrangian, we can set $\theta = 0$. However, we know that the CKM matrix is complex, and thus CP symmetry must be spontaneously broken. We need to avoid the generation of $\bar{\theta}$ from the spontaneous CP violation. The following papers are useful to understand this section.

- Dine-Draper [69]

8.1 BBP model

	$SU(3)_C$	$SU(2)_L$	$U(1)_Y$	Z_2
D_L	3	1	$-1/3$	$-$
D_R	3	1	$-1/3$	$-$
Φ	1	1	0	$-$

Table 8. Matter content of BBP model.

The minimal model of Nelson-Barr mechanism is BBP model [70]. In this model, we introduce a vector-like quark $D_{L,R}$ and a complex scalar Φ. D is a $SU(2)_L$ singlet with hypercharge $-1/3$. We impose Z_2 symmetry as

$$\Phi \to -\Phi, \qquad D_{L,R} \to -D_{L,R}, \tag{8.1}$$

and all of the SM fields are Z_2 even. The Lagrangian of this model is given by

$$\mathcal{L} = -\bar{u}_R y_u \tilde{H}^\dagger q_L - \begin{pmatrix} \bar{d}_R & \bar{D}_R \end{pmatrix} \begin{pmatrix} y_d H^\dagger & \kappa\Phi + \kappa'\Phi^* \\ 0 & M_D \end{pmatrix} \begin{pmatrix} q_L \\ D_L \end{pmatrix} + h.c., \tag{8.2}$$

We impose CP symmetry on the Lagrangian, and all of the parameters, i.e., y_u, y_d, κ, κ', and M_D, are real. Then, we assume that Φ obtains a complex VEV as $\langle\Phi\rangle \neq \langle\Phi\rangle^*$.

Let us define

$$m_u \equiv y_u\langle H\rangle, \qquad m_d \equiv y_d\langle H\rangle, \qquad B \equiv \kappa\langle\Phi\rangle + \kappa'\langle\Phi\rangle^*, \qquad \mathcal{M} \equiv \begin{pmatrix} m_d & B \\ 0 & M_D \end{pmatrix} \tag{8.3}$$

Note that B, κ, and κ' are three-component vectors in the flavor space of down-type quarks. Then, we can easily see that

$$\arg\det m_u = 0, \qquad \arg\det\mathcal{M} = 0. \tag{8.4}$$

Thus, we obtain

$$\bar{\theta} = 0 \tag{8.5}$$

at tree level.

Let us discuss the CKM matrix in this model. The mass matrix of up-type quark is diagonalized as

$$O_{u_R}^T m_u O_{u_L} = \hat{m}_u. \tag{8.6}$$

where O_{u_L} and O_{u_R} are real orthogonal matrices. The mass matrix of down-type quark is diagonalized as

$$\mathcal{U}_R^\dagger \mathcal{M} \mathcal{U}_L = \begin{pmatrix} \hat{m}_d & 0 \\ 0 & \hat{M}_D \end{pmatrix} \tag{8.7}$$

From eq. (8.3), we obtain

$$\mathcal{M}^\dagger \mathcal{M} = \begin{pmatrix} m_d^T m_d & m_d^T B \\ B^\dagger m_d & M_D^2 + B^\dagger B \end{pmatrix} \tag{8.8}$$

In the limit of $m_d \ll B, M_D$, the mass eigenvalue of heavy quark is obtained as $\hat{M}_D \simeq \sqrt{M_D^2 + B^\dagger B}$. $\mathcal{U}_L$ is given by

$$\mathcal{U}_L = \begin{pmatrix} \mathbf{1}_3 & m_d^T B/\hat{M}_D^2 \\ -B^\dagger m_d/\hat{M}_D^2 & 1 \end{pmatrix} \begin{pmatrix} U_{d_L} & \\ & 1 \end{pmatrix} + \mathcal{O}(m_b^2/\hat{M}_D^2). \tag{8.9}$$

where U_{d_L} is a 3×3 unitary matrix which diagonalizes the mass matrix of down-type quark as

$$U_{d_L}^\dagger \left(m_d^T m_d - \frac{m_d^T B B^\dagger m_d}{\hat{M}_D^2} \right) U_{d_L} = \hat{m}_d^2. \tag{8.10}$$

The CKM matrix is given by

$$V_{\text{CKM}} = O_{u_L}^\dagger U_{d_L}. \tag{8.11}$$

Thus, V_{CKM} can be complex! To obtain $\mathcal{O}(1)$ complex phase, $|B|$ and M_D should be of the same order.

To summarize, in BBP model, we can obtain $\bar{\theta} = 0$ at tree level because m_u is real and also $\det\{\mathcal{M}\}$ is real. However, $\mathcal{M}$ has complex off-diagonal elements! As a result, we obtain a complex CKM matrix from this mass matrix.

8.2 Quality problem in Nelson-Barr mechanism

We have seen that $\bar{\theta} = 0$ is realized in BBP model at tree level. However, $\bar{\theta}$ can be generated by radiative corrections. Here we discuss:

- One-loop diagram with Higgs portal coupling
- Two-loop 'dead duck' diagram
- Dimension-5 operators

For a review, see, e.g., [69].

One-loop diagram with Higgs portal coupling

The following Higgs portal coupling of Φ is also allowed under Z_2 symmetry:

$$\mathcal{L} \ni \lambda \Phi^2 |H|^2 + \text{h.c.}. \tag{8.12}$$

This coupling induces one-loop correction to the Yukawa coupling as [69, 70]

$$\delta y_{d,ij} \sim \frac{\lambda y_{d,kj} \kappa'^*_k \kappa^*_i \langle \Phi \rangle^2}{16\pi^2 M_\Phi^2}. \tag{8.13}$$

Then, we obtain $\delta\bar{\theta}$:

$$\delta\bar{\theta} = \text{Imtr}[y_d^{-1} \delta y_d] \sim \frac{\lambda \kappa \kappa' \langle \Phi \rangle^2}{16\pi^2 M_\Phi^2}. \tag{8.14}$$

By using $M_D \sim |\kappa| \langle \Phi \rangle$, we obtain the upper bound on λ as

$$\lambda \lesssim 10^{-8} \times \frac{M_\Phi^2}{M_D^2}. \tag{8.15}$$

Note that a natural value of λ is of the order of $\langle H \rangle^2 / \langle \Phi \rangle^2$ to realize $\langle H \rangle \ll \langle \Phi \rangle$ [70]. Thus, this correction decouples by assuming $\langle H \rangle \ll \langle \Phi \rangle$.

Two-loop 'dead duck' diagram

Non-decoupling effect on $\delta\bar{\theta}$ comes from the self-interaction of Φ:

$$\mathcal{L} \ni -\gamma \Phi^3 \Phi^* + \text{h.c.}. \tag{8.16}$$

Two-loop 'dead duck' diagram [66] gives a correction to the mass of vector-like quark:

$$\delta M_D \sim M_D \frac{g_s^2 \gamma}{(16\pi^2)^2} \frac{\kappa^2 \langle \Phi \rangle^2}{\max[M_D^2, M_\Phi^2]} \tag{8.17}$$

Thus, we obtain $\delta\bar{\theta}$ as

$$\delta\bar{\theta} = \frac{\text{Im}[\delta M_D]}{M_D} \sim \frac{g_s^2 \gamma}{(16\pi^2)^2} \frac{\kappa^2 \langle \Phi \rangle^2}{\max[M_D^2, M_\Phi^2]}. \tag{8.18}$$

By using $M_D \sim \kappa \langle \Phi \rangle$, we obtain the upper bound on γ as

$$\gamma \lesssim 10^{-6} \times \max\left[\frac{M_\Phi^2}{M_D^2}, 1\right]. \tag{8.19}$$

This effect does not decouple in the limit of a large Nelson–Barr scale. In general, $\gamma \sim \mathcal{O}(1)$ and this requires $M_D/M_\Phi \lesssim 10^{-3}$ or equivalently $\kappa \lesssim 10^{-3}$ [71].

Dimension-5 operators

In addition to radiative corrections, higher dimensional effective interactions suppressed by the cutoff scale of the theory can also induce $\delta\bar{\theta}$. The following dimension-5 effective interactions are allowed by the Z_2 symmetry:

$$\mathcal{L} \sim \frac{1}{\Lambda}\Phi^2 \bar{D}_R D_L + \frac{1}{\Lambda}\Phi \bar{D}_R H^\dagger q_L, \tag{8.20}$$

where Λ is a cutoff scale of the theory. These terms break the structure of the quark mass matrix. As a result, we obtain the correction to $\bar{\theta}$ as [72]

$$\delta\bar{\theta} \sim \frac{\langle\Phi\rangle^2}{M_D\Lambda} + \frac{B_i(y_d)^{-1}_{ij}\langle\Phi\rangle}{M_D\Lambda} \tag{8.21}$$

By using $M_D \sim |\kappa||\langle\Phi\rangle| = |B|$, we obtain $\delta\bar{\theta}$ as

$$\delta\bar{\theta} \sim \frac{\langle\Phi\rangle}{\Lambda} \times \frac{1}{\min[y,\kappa]}. \tag{8.22}$$

By assuming $\Lambda \lesssim M_{\rm pl}$ and $\delta\bar{\theta} \lesssim 10^{-10}$, we obtain

$$\langle\Phi\rangle \lesssim 10^8 \text{ GeV} \times \min[y,\kappa]. \tag{8.23}$$

This gives a severe upper bound on the Nelson–Barr energy scale. See also, refs. [69, 71].

Solutions to the quality problem

To solve the strong CP problem by the Nelson–Barr mechanism, we need to suppress the dangerous corrections $\delta\bar{\theta}$ described so far. Let us briefly summarize the existing proposals to address this problem.

In general, eq. (8.19) and eq. (8.23) give severe constraints on Nelson–Barr models [71, 72]. First, the constraint in eq. (8.23) with $y_{u/d} \sim 10^{-5}$ requires $\langle\Phi\rangle \lesssim 10^3$ GeV. In addition, $\gamma \sim \mathcal{O}(1)$ with eq. (8.19) requires $\kappa \lesssim 10^{-3}$ and this leads to $M_D \lesssim 1$ GeV! Apparently, this destroys the unitarity of the CKM matrix, and such a light exotic colored particle has not been observed at collider experiments. Viable models can be constructed, for example, by assuming an extra symmetry [71–74], a five-dimensional model [75], and CP breaking mediation by higher dimensional operators [76, 77].

Supersymmetry (SUSY) is one of the simplest solutions to solve the quality problem. Since the interaction terms in eqs. (8.12), (8.16), and (8.20) have a non-holomorphic dependence on Φ, their structure can be naturally controlled [78]. Since additional sources of $\bar{\theta}$, such as the gluino mass, can appear in SUSY models, gauge mediation is one of the favored SUSY-breaking mediation mechanisms to avoid this issue [69, 79–81].

8.3 Summary of this section

In this section, we have discussed the Nelson–Barr mechanism, which is another solution to the strong CP problem based on spontaneous CP violation. In this mechanism, we impose CP symmetry on the Lagrangian and assume that CP symmetry is spontaneously broken. Then, we can obtain $\bar{\theta} = 0$ at tree level. However, radiative corrections and higher dimensional operators can generate $\bar{\theta} \neq 0$ in general. This requires more dedicated model building to suppress the dangerous corrections to $\bar{\theta}$.

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A η' mass in large N_c QCD: Witten-Veneziano relation

In the large N_c limit, the relation between the $U(1)_A$ problem and the topological susceptibility becomes transparent [82, 83]. The topological susceptibility $\chi(0)$ can be expanded in $1/N_c$ as

$$\chi(q^2) = \chi_{\rm YM}(q^2) + \frac{N_f}{N_c}\chi_q(q^2) + \mathcal{O}(N_c^{-2}). \tag{A.1}$$

Here $\chi_{\rm YM}(q^2)$ is the pure gluonic contribution, which is the same as pure $SU(N_c)$ Yang-Mills theory. $\chi_q(q^2)$ is the contribution from quark loops. In chiral limit ($m_q = 0$), $\chi(0) = 0$ should be satisfied. However, gluonic contribution starts from $\mathcal{O}(N_c^0)$, and quark contribution starts from $\mathcal{O}(N_c^{-1})$. How can $\chi(0) = 0$ be satisfied?

Witten and Veneziano [82, 83] pointed out that this puzzle can be resolved if $\chi_q(q^2)$ has a pole as

$$\chi_q(q^2) = \frac{a^2}{q^2 - m_{\eta'}^2}, \tag{A.2}$$

and $m_{\eta'}^2 \propto 1/N_c$. By using $\chi(0) = 0$, we obtain

$$m_{\eta'}^2 = \frac{N_f}{N_c}\frac{a^2}{\chi_{\rm YM}(0)}. \tag{A.3}$$

Then,

$$a = \sqrt{\frac{N_c}{N_f}} \times \langle 0|\frac{g_s^2}{32\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu}|\eta'\rangle = \sqrt{\frac{N_c}{N_f}}\frac{1}{2N_f}\langle 0|\partial_\mu J^\mu_A|\eta'\rangle = \sqrt{\frac{N_c}{N_f}}\frac{1}{2N_f}\sqrt{2N_f}fm_{\eta'}^2 \tag{A.4}$$

Then, we obtain [82, 83]

$$m_{\eta'}^2 = \frac{1}{2N_f}\frac{f^2 m_{\eta'}^4}{\chi_{\rm YM}(0)} = \frac{2N_f}{f^2}\chi_{\rm YM}(0). \tag{A.5}$$

A.1 Topological susceptibility in chiral Lagrangian

Finally, let us derive $\chi(0)$ from the chiral Lagrangian.

$$V = -f_\pi^2 B_0 \sum_q m_q \cos(\theta_q + \varphi_q) + \frac{1}{2}\chi_{\rm YM}\left(\sum_q \varphi_q\right)^2 \tag{A.6}$$

$$\chi = \left(\frac{1}{\chi_{\rm YM}} + \frac{1}{f_\pi^2 B_0 m_*}\right)^{-1} \tag{A.7}$$

Adding this effect into the chiral Lagrangian, we obtain the following mass relation :

$$m_{\eta'}^2 = \frac{6}{f_\pi^2}\chi_{\rm YM} + 2m_K^2 - m_\eta^2. \tag{A.8}$$

See, e.g., ref. [38]. Thus, nonzero topological susceptibility is closely connected to the mass of η'!

A.2 Note on Coleman's textbook

Coleman [84] says "*I emphasize that there is no connection between the eta and the U(1) problem. The eta is a red herring; it is just another hadron; it is no more a relic of a U(1) Goldstone boson than is the N**.*" But this lecture note is written in 1977. Witten and Veneziano's papers are published in 1979. So, at the time of writing this textbook, the connection between η' and $U(1)_A$ problem was not clear yet. After Witten and Veneziano's papers, it becomes clear that η' is closely connected to $U(1)_A$ problem.

B Radiative correction of $\bar{\theta}$ in the Standard Model

Suppose that $\bar{\theta} = 0$ at some high energy scale. Can this explain the smallness of neutron EDM? To discuss this, we need to discuss the running of $\bar{\theta}$ and finite correction to $\bar{\theta}$.

B.1 Running of $\bar{\theta}$

Let us discuss the running of $\arg\det y_u y_d$ [85]. Loop diagram contribution is given as

$$\mathrm{Im}\left[\mathrm{tr}\left(\prod_{i=1}^{k} m_u^{2n_i} V_{\rm CKM} m_d^{2m_i} V_{\rm CKM}^\dagger\right)\right]. \tag{B.1}$$

For Hermitian matrices A and B, $\mathrm{Im}[\mathrm{tr}(AB)] = 0$.

Thus, for $k = 1$,

$$\mathrm{Im}\left[\mathrm{tr}\left(m_u^{2n_1} V_{\rm CKM} m_d^{2m_1} V_{\rm CKM}^\dagger\right)\right] = 0, \tag{B.2}$$

and there is no contributions.

For $k = 2$, we obtain

$$\mathrm{Im}\left[\mathrm{tr}\left(m_u^2 V_{\rm CKM} m_d^2 V_{\rm CKM}^\dagger m_u^2 V_{\rm CKM} m_d^2 V_{\rm CKM}^\dagger\right)\right] = 0, \tag{B.3}$$

$$\mathrm{Im}\left[\mathrm{tr}\left(m_u^4 V_{\rm CKM} m_d^2 V_{\rm CKM}^\dagger m_u^2 V_{\rm CKM} m_d^2 V_{\rm CKM}^\dagger\right)\right] = 0, \tag{B.4}$$

$$\mathrm{Im}\left[\mathrm{tr}\left(m_u^2 V_{\rm CKM} m_d^4 V_{\rm CKM}^\dagger m_u^2 V_{\rm CKM} m_d^2 V_{\rm CKM}^\dagger\right)\right] = 0, \tag{B.5}$$

and, at the leading order, we obtain nonzero contribution as

$$\mathrm{Im}\left[\mathrm{tr}\left(m_u^4 V_{\rm CKM} m_d^4 V_{\rm CKM}^\dagger m_u^2 V_{\rm CKM} m_d^2 V_{\rm CKM}^\dagger\right)\right] \simeq m_t^4 m_b^4 m_c^2 m_s^2 J. \tag{B.6}$$

This corredponds to 7-loop diagram [85].

B.2 Finite correction to $\bar{\theta}$

§2.2 in [86], §4.1 in [39].

One-loop strong penguin diagram gives CP violating four quark operator. One-loop pion loop (two-loop in total) diagram with this operator gives leading contribution to d_n as [87–89]

$$d_n \sim 10^{-32}\ e\ \mathrm{cm}, \tag{B.7}$$

C Miscellaneous

C.1 Anthropic principle?

The strong CP problem is a naturalness problem. For a review of naturalness, see, e.g., [90]. There is severe naturalness problems in the Standard Model, such as

- electroweak hierarchy problem
- cosmological constant problem

Should we really care about the smallness of $\bar{\theta}$? One of specific feature of the strong CP problem is that it is very difficult to solve this problem by anthropic principle. How nonzero $\bar{\theta} \gtrsim 10^{-10}$ could affect the life? There have been some attempts but it is not clear the strong CP problem can be solved by the anthropic principle [91, 92].

C.2 Solution within QCD?

There have been several trials to solve the strong CP problem *within* QCD. For recent trials, see, for example, [93–96], [97–99], [100–102], [103].

However, this direction is extremely difficult. As we have seen, the strong CP problem is closely conntected to $U(1)_A$ problem [26, 38]. See also [104] [105].

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